Peptide Protocol Reference

Reconstitution, protocols, cycles, and safety.

Last updated: May 16, 2026

What's inside

Glossary + Monitoring — Acronyms, recon math, lab ranges.
Universal Safety — Contraindications, lab cadence, and stop criteria that apply to every peptide.
Table of Contents — Grouped by mechanism. Popular Compounds list beneath.
Peptide pages (47) — Alphabetical. Reference + Safety on one page when both fit, otherwise two.
Sources & References — Full grouped source list at the end of the doc.

How to read each peptide page

Reference — Reconstitution, Protocol, Cycle length, Timing, Injection site, Storage, Side effects, Pairs well with / Avoid, References.
Safety — Contraindications, Warnings, Drug interactions, Pregnancy/Lactation, Labs, Stop criteria, Patient education.
Links — Blue underlined peptide names jump to the linked peptide. Reference URLs open in a browser.

Reference sites used

Researched and cross-audited across **45+ sources** — community DBs, compounding pharmacies, FDA labels, PubMed/PMC, and clinical trial registries. Full grouped list in the **Sources & References** appendix at the end.

Each peptide page links the 6 most-applicable sites for that compound.

Groups

	Fat loss and metabolism	9 compounds
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47 compounds

Informational only. *Not medical advice. Consult a qualified clinician before beginning any protocol.*

Glossary and Lab Reference

Acronyms used throughout the guide and adult lab reference ranges.

Acronyms and glossary

SubQ / SC Subcutaneous (under the skin, fatty layer)	**IM** Intramuscular	**IV** Intravenous
ID Intradermal	**BAC** Bacteriostatic water (sterile water + 0.9% benzyl alcohol)	**mcg / µg** Microgram (1/1000 of a milligram)
IU International unit	**U-100** 100-unit/mL insulin syringe (1u = 0.01 mL)	**GHRH** Growth hormone releasing hormone (CJC-1295, Tesamorelin)
GHRP Growth hormone releasing peptide (Ipamorelin)	**GLP-1** Glucagon-like peptide-1 (Semaglutide)	**GIP** Glucose-dependent insulinotropic polypeptide
DPP-4 Dipeptidyl peptidase-4 (degrades GLP-1; inhibitors prolong it)	**TRT / HRT** Testosterone / hormone replacement therapy	**PCT** Post-cycle therapy (recovery after suppressive cycle)
HPG axis Hypothalamic-pituitary-gonadal axis	**HPA axis** Hypothalamic-pituitary-adrenal axis	**MTC** Medullary thyroid carcinoma
MEN 2 Multiple endocrine neoplasia type 2 (hereditary cancer syndrome)	**BRCA** BRCA1 / BRCA2 hereditary breast/ovarian cancer syndromes	**Lynch** Lynch syndrome (hereditary colorectal cancer)
Li-Fraumeni Hereditary multi-cancer syndrome (TP53 mutation)	**USPSTF** US Preventive Services Task Force (cancer screening recommendations)	**AKI** Acute kidney injury
CKD Chronic kidney disease	**OHSS** Ovarian hyperstimulation syndrome	**MI** Myocardial infarction (heart attack)
T2DM Type 2 diabetes mellitus	**IBD** Inflammatory bowel disease (Crohn's, ulcerative colitis)	**GERD** Gastroesophageal reflux disease
SCFE Slipped capital femoral epiphysis (pediatric hip)	**CIRS** Chronic inflammatory response syndrome	**IGF-1** Insulin-like growth factor 1
IGFBP IGF binding protein (prolongs IGF-1 half-life)	**BDNF** Brain-derived neurotrophic factor	**NGF** Nerve growth factor
VEGF Vascular endothelial growth factor	**TNF-α** Tumor necrosis factor alpha (pro-inflammatory cytokine)	**AMPK** AMP-activated protein kinase (energy sensor)
NNMT Nicotinamide N-methyltransferase	**NAD+** Nicotinamide adenine dinucleotide	**TMAO** Trimethylamine N-oxide (gut metabolite, CV concern)
BMP Basic metabolic panel (glucose, electrolytes, BUN, creatinine)	**CMP** Comprehensive metabolic panel (BMP + liver enzymes + albumin)	**CBC** Complete blood count
AST / ALT Liver enzymes (aspartate / alanine aminotransferase)	**BUN** Blood urea nitrogen (kidney marker)	**eGFR** Estimated glomerular filtration rate (kidney function)
TSH / fT4 Thyroid stimulating hormone / free thyroxine	**LH / FSH** Luteinizing hormone / follicle-stimulating hormone	**SHBG** Sex hormone binding globulin
PSA Prostate-specific antigen	**hsCRP** High-sensitivity C-reactive protein (inflammation)	**MMA** Methylmalonic acid (B12 deficiency marker)
ULN Upper limit of normal (target ceiling for IGF-1 etc.)	**HbA1c** Glycated hemoglobin (3-month glucose average)	**CD3 / CD4 / CD8 / NK** T-cell subsets and natural killer cells (immune panel)
NSAID Non-steroidal anti-inflammatory drug (ibuprofen, naproxen, etc.)	**DOAC** Direct oral anticoagulant (apixaban, rivaroxaban, dabigatran)	**SSRI / SNRI** Selective serotonin / serotonin-norepinephrine reuptake inhibitors
MAOI Monoamine oxidase inhibitor	**PDE5** Phosphodiesterase 5 inhibitor (sildenafil, tadalafil)	**AI** Aromatase inhibitor (anastrozole, letrozole)
SERM Selective estrogen receptor modulator (clomiphene, tamoxifen)	**GnRH** Gonadotropin-releasing hormone (triptorelin, leuprolide)	**CYP3A4** Cytochrome P450 3A4 (major drug-metabolizing enzyme)
ABCDE Mole monitoring criteria: Asymmetry, Border, Color, Diameter, Evolution	**FDA** US Food and Drug Administration	**EMA** European Medicines Agency
NMPA National Medical Products Administration (China)	**USPI** US prescribing information (FDA label)	**EPAR** European public assessment report (EMA equivalent of USPI)
ASA American Society of Anesthesiologists (anesthesia guidance)	**ICH-GCP** International Council for Harmonisation good clinical practice	**RCT** Randomized controlled trial
PK / PD Pharmacokinetics (drug movement) / pharmacodynamics (drug effect)	**t½** Half-life (time for plasma concentration to fall by 50%)	**Tmax** Time to peak plasma concentration
COA Certificate of analysis (purity / potency lab report)	**503A / 503B** FDA compounding pharmacy regulatory categories	

Monitoring schedule by class

Class-level cadence for the most-monitored labs. Compound-specific markers are listed on each peptide's safety page.

Class / examples	Baseline	Wks 4-8	Wk 12 / End	Quarterly during chronic use
Class A — GLP-1 / dual / triple agonists *(Semaglutide, Tirzepatide, Retatrutide)*	BMP/CMP, CBC, HbA1c, lipid panel, TSH, β-hCG (women)	BMP at every dose escalation; lipase if abdominal pain	HbA1c, lipid panel, repeat eGFR	HbA1c q3 mo; BMP q6 mo; eye exam yearly
Class B — GH secretagogues *(CJC-1295 No DAC, Tesamorelin, Ipamorelin)*	IGF-1, fasting glucose, HbA1c, lipid panel, age-appropriate cancer screening complete	Fasting glucose Wk 4-8	IGF-1, fasting glucose, HbA1c, lipid panel	IGF-1 q3 mo (target ≤ULN for age)
Class C — Melanocortin agonists *(MT-1, MT-2, PT-141)*	Dermatology mole map (photographic), BP log	BP before each dose	Dermatology re-check at 12 weeks during active use	Dermatology q6-12 mo; BP log on PT-141
Class D — Khavinson short peptides *(Epithalon, Pinealon)*	CBC, CMP; tissue-specific marker (salivary melatonin for Epithalon)	—	Repeat tissue-specific marker after 10-day course	Repeat course 2-3×/year; no continuous monitoring required
Class E — Russian nootropics *(Selank, Semax, intranasal variants)*	CBC, CMP	—	—	No specific labs; subjective cognitive/mood follow-up
Class F — HPG axis (gonadotropins, GnRH) *(HCG, Kisspeptin-10/54)*	Total + free T, LH, FSH, sensitive E2, SHBG, prolactin, CBC + Hct, lipid panel, PSA (men 40+), semen analysis (fertility)	E2 + Hct Wk 4-6	Total + free T, E2, semen analysis (fertility)	Hct q3 mo; E2 q3 mo; PSA per age guidelines; D-dimer if clotting concern
Class G — Lipotropic blends *(Lipo-C, LC120/216/526, MIC Blend, HHB, SHB, L-Carnitine)*	B12, folate, MMA, homocysteine, TSH, liver enzymes, Hct	—	B12, homocysteine, liver enzymes	B12 + homocysteine q3-6 mo on chronic use
Class H — Recombinant proteins *(HGH/Somatropin, IGF-1, IGF-1 LR3)*	IGF-1, fasting glucose, HbA1c, TSH/fT4, AM cortisol, lipid panel, age-appropriate cancer screening complete	Fasting glucose Wk 4-8	IGF-1, fasting glucose, HbA1c	IGF-1 q3 mo (target ≤ULN for age); HbA1c q3 mo; TSH/fT4 q6 mo

Universal Safety

These rules apply to every peptide in this reference. Compound-specific contraindications and warnings are listed on each peptide's own page.

Contraindications (apply to all peptides)

- Known hypersensitivity to the active compound or any excipient (BAC water benzyl alcohol, mannitol, etc.)
- Active systemic infection at injection sites
- Pregnancy and lactation unless specifically indicated and supervised by a clinician (most peptides have no pregnancy safety data)
- Pediatric use unless specifically indicated (most peptides have no pediatric data)

Stop and seek medical care for

- Signs of anaphylaxis: throat tightness, difficulty breathing, hives, swelling of face/lips/tongue. Seek emergency care
- Signs of serious infection at injection site: spreading redness beyond the puncture, fever, purulent drainage
- New unexplained symptoms during a cycle that don't have another clear cause

Injection technique and storage basics

- Aseptic technique: alcohol swab the vial septum and the injection site, use a single-use sterile needle for every injection, never share vials or needles between people
- Sharps disposal in an FDA-cleared sharps container, never household trash
- Cold chain: if a refrigerated peptide reaches room temperature for more than the labeled stability window, discard it
- Mark the date of first puncture on every multi-dose vial
- Inspect every vial before injection: clear solution (unless labeled otherwise), no particulates, no cloudiness, no color change
- Rotate injection sites to avoid lipoatrophy, induration, and infection
- Carry a record of what you're injecting if you need emergency care

GH-axis cancer screening (required before any GH/IGF-1 peptide)

Applies to: CJC-1295 family, Ipamorelin, Tesamorelin, HGH (Somatropin), IGF-1 family. Same screening logic applies to any GH or IGF-1 axis compound.

GH/IGF-1 axis stimulation accelerates the growth of cells that respond to it - including any malignancy you don't know about. Before any first cycle, complete age-appropriate cancer screening: colonoscopy (USPSTF 45+), mammogram (women 40+), cervical screening, prostate exam + PSA (men 50+, or 45+ with risk), full-body dermatology check, lung CT if smoking history, full symptom review (unexplained weight loss, persistent cough, blood in stool/urine, persistent abdominal pain, lumps, night sweats, persistent fatigue). Family history red flags warranting additional screening: first-degree relative with cancer before age 50, multiple cancers in family, known hereditary cancer syndromes (BRCA, Lynch, MEN, Li-Fraumeni). If you can't or won't get age-appropriate cancer screening before starting, you shouldn't start. Repeat IGF-1 every cycle; if IGF-1 rises above the upper limit of normal for your age, drop the dose.

Recommended baseline labs (any cycle start)

- BMP or CMP (creatinine, eGFR, electrolytes, glucose; CMP adds liver enzymes which is preferred)
- CBC with differential
- HbA1c if any metabolic risk
- Lipid panel
- TSH
- Pregnancy test in females of childbearing potential before starting any peptide cycle
- Urinalysis is reasonable but not mandatory

Note: *hsCRP and routine calcitonin are NOT standard pre-cycle labs. hsCRP is not actionable for peptide monitoring. Per FDA Ozempic label, routine calcitonin monitoring is of "uncertain value" for early MTC detection.*

Lab reference ranges (US adult)

Orientation only — your reference lab may differ by 5-15%. Age- and sex-specific ranges apply.

Marker	Normal range	Notes
IGF-1	Age-dependent (peaks 20s, declines ~1%/yr)	20s: 200-400 ng/mL; 50s: 100-250; 70s: 70-180. Stay within YOUR age range, not a young range.
Fasting glucose	70-99 mg/dL	Pre-diabetes: 100-125. Diabetes: ≥126. GLP-1 risk monitoring.
HbA1c	<5.7%	Pre-diabetes 5.7-6.4%, diabetes ≥6.5%. Monitor q3 months on GLP-1s.
eGFR	>=60 mL/min/1.73 m²	<60 = CKD; <30 = severe. Hold GLP-1s if eGFR drops >20% from baseline.
Total testosterone (men)	300-1000 ng/dL	<300 generally treated. Free T and SHBG more useful clinically.
Estradiol (men, sensitive E2)	10-40 pg/mL	Above 40 with HCG/HMG often warrants AI consideration.
Hematocrit	Men 41-53%; Women 36-46%	>54% = polycythemia; donate blood or hold therapy.
LDL cholesterol	<100 mg/dL optimal	<70 if cardiac risk. <130 acceptable for most.
hsCRP	<3 mg/L	Inflammation marker. >3 = elevated. Not a routine peptide-monitoring lab.
TSH	0.4-4.5 mIU/L	Hypothyroid: >4.5. Hyperthyroid: <0.4. Re-check 6-8 wk after dose changes.
B12	200-900 pg/mL	Target 400-900 for symptom prevention. GLP-1s may deplete.
Vitamin D (25-OH)	30-100 ng/mL	<20 deficiency, 20-30 insufficient. Aim 40-60.

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Grouped by mechanism below. Body pages are alphabetical, so jumping A→Z is also easy. Click any entry to navigate.

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Popular Compounds & Blends

Community popularity ranking. Click any name to jump to its page.

1. CJC-1295 (No DAC) + Ipamorelin	Boosts natural growth-hormone pulses; injected pre-bed
2. GHK-Cu	Skin, hair, and connective-tissue repair
3. GLOW Blend	BPC + TB + GHK-Cu — for skin, hair, post-surgical recovery, wound healing
4. Glutathione	Master antioxidant; supports liver detox
5. HCG	Preserves natural testosterone during TRT; supports fertility
6. HGH (Somatropin)	Direct growth hormone — FDA-approved; used off-label by adults
7. KLOW Blend	GLOW + KPV — adds gut and joint anti-inflammatory coverage for IBD, Crohn's, colitis
8. MOTS-c	Mitochondrial peptide for energy and metabolic support; pairs with NAD+
9. NAD+	Cellular energy support — declines with age
10. PT-141 (Bremelanotide)	Libido enhancement — FDA Vyleesi
11. Retatrutide	Strongest weight-loss compound in human trials
12. Tesamorelin	Targets visceral (belly) fat — FDA Egrifta
13. Wolverine Blend	Wolverine Blend — most popular repair pairing; BPC + TB pre-mixed in one vial

5-AMINO-1MQ

FAT LOSS AND METABOLISM

In plain English	*Stops your body from burning through its own energy fuel so fat-burning runs more efficiently.*
What it does	Blocks NNMT enzyme to support fat metabolism, cellular energy, and NAD+ preservation. Inhibits Nicotinamide N-methyltransferase (NNMT), reducing NAD+ consumption and increasing SIRT1/AMPK activity.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 0.25 mL	20 mg/mL	1 unit = 0.20 mg
10 mg	10mg + 0.50 mL	20 mg/mL	1 unit = 0.20 mg
50 mg	50mg + 2.5 mL	20 mg/mL	1 unit = 0.20 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-2	Daily AM	**2.5 mg**	12.5u	Tolerance check. SC abdomen or thigh. Assess GI tolerance and stimulation before stepping up.
Weeks 3-8	Daily or 2.5 mg twice a day	**5 mg**	25u	Working dose. Half-life 3.8-6.9 h, so splitting into two daily doses gives steadier NNMT inhibition. Hold here if Weeks 1-2 were comfortable.
Off	4 weeks	-	-	Standard cycle: 8 weeks on, 4 weeks off. Extended: 12 on, 6 off.
Cycle length	6-8 weeks on, 2-4 weeks off.			
Timing	AM only — stimulating; avoid evening doses (sleep disruption)			

Handling & Stacking

Injection site	Abdomen or outer thigh.
Storage	Lyophilised: -4°F (up to 24 months). Reconstituted: 36-46°F, use within 4 weeks. No freeze-thaw.
Side effects	Appetite suppression, mild stimulation, injection-site stinging.
What to expect	Mild appetite suppression and slight stimulation within 1-2 hours. Fat loss is gradual - most noticeable at 6-8 weeks with consistent caloric deficit.
Supply math	**5mg** → 48 vials | **10mg** → 24 vials | **50mg** → 5 vials *(8-wk @ 5 mg × 7/wk)*
Pairs well with	**NAD+** (preserves injected **NAD+** via NNMT inhibition); **MOTS-c** (overlapping metabolic pathways); **Tirzepatide** / **Semaglutide** (NNMT + GLP-1 synergy)
Avoid	Metformin or Berberine (additive AMPK activation risks significant glucose drops)
References	peptidedosingprotocols · researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• No human safety data; pregnancy, lactation, pediatrics excluded. • Active malignancy (NAD+ pathway interactions theoretical).
Warnings	• No human RCT data; community use only. • NNMT inhibition broadly affects methylation. Long-term safety unknown.
Drug interactions	• Metformin and berberine (additive AMPK activation, glucose drops) — hard contraindication during the same dosing window
Labs	• Compound-specific: Fasting glucose (Week 0/4/8)
Stop criteria	• Persistent hypoglycemia, unexplained fatigue, mood changes
Patient education	• Research compound. No published human RCTs. NAD+ pathway effects are theoretical based on cell and animal data.

ACETIC ACID

SUPPLEMENTS AND COFACTORS

In plain English	*A diluent used to mix up IGF-1 family peptides. It's a solvent, not a peptide. Don't inject it on its own.*
What it does	0.6% sterile acetic acid solution - the required solvent for IGF-1 LR3 and similar peptides that don't dissolve in BAC water. IGF-1 family peptides have isoelectric points near neutral pH and aggregate or precipitate in BAC water.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
3 ml / 5 ml	Pre-mixed solution	0.6% acetic acid solution	Not for injection on its own

Handling & Stacking

Injection site	Solvent only - never inject undiluted acetic acid. Reconstitute peptide first, then dilute with sterile saline 1:1 immediately before injection.
Storage	Room temperature. Single-use ampoules preferred over multi-dose vials. Check for cloudiness before use.
Side effects	Injection-site sting (acidic pH). Tissue irritation if injected undiluted.
What to expect	Slight sting at the injection site is normal due to the acidic pH. Always dilute with sterile saline before injecting if possible.
Pairs well with	**IGF-1 LR3**, **IGF-1 (Standard)** - required solvent. Sterile saline - dilute 1:1 before injection.
Avoid	Direct injection without dilution. Peptides that prefer BAC water (almost all peptides except IGF-1 family).
References	peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Warnings	• Solvent only; never inject undiluted (causes burning, tissue irritation)
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Patient education	• Always dilute with sterile saline before injection

BACTERIOSTATIC WATER

SUPPLEMENTS AND COFACTORS

In plain English	*Sterile water with a tiny amount of preservative (0.9% benzyl alcohol). This is what most peptides get mixed with. Good for 28 days once you've punctured the vial.*
What it does	Sterile water with 0.9% benzyl alcohol preservative - the standard solvent for reconstituting most lyophilised peptides. Benzyl alcohol at 0.9% is bacteriostatic (prevents bacterial multiplication) without being cytotoxic at the volumes used.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 ml / 30 ml	Pre-mixed solution	0.9% benzyl alcohol in sterile water	Not for injection on its own

Handling & Stacking

Injection site	N/A - solvent only.
Storage	Room temperature unopened. After first puncture: room temp OK; refrigerated extends use to ~30 days. Cap sterile and label puncture date.
Side effects	Benzyl alcohol allergy is rare but documented; switch to preservative-free sterile water if confirmed. Contraindicated in neonates due to gasping syndrome from benzyl alcohol toxicity.
What to expect	Clear, colourless solution with no preservative odour. Should pour cleanly without bubbles or particulates. Mark the date of first puncture on the vial.
Pairs well with	Almost every lyophilised peptide. Insulin syringe (U-100) for accurate volume transfer.
Avoid	**IGF-1 LR3** family (use 0.6% **acetic acid**). Neonates (benzyl alcohol toxicity). Patients with confirmed benzyl alcohol allergy (use preservative-free sterile water for single-use only).
References	peptidedosages · DailyMed · PubMed · ClinicalTrials · Drugs.com

Safety & Patient Education

Contraindications	• Benzyl alcohol allergy, neonates (benzyl alcohol toxicity)
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Patient education	• Mark first puncture date; discard at 28-30 days

BPC-157

TISSUE REPAIR AND RECOVERY

In plain English	*The go-to recovery peptide. Speeds up healing in tendons, muscles, ligaments, and gut lining.*
What it does	Research-phase peptide; no published human RCTs. Community off-label use. Stimulates angiogenesis via VEGF, activates GH receptor on tendon fibroblasts, promotes nitric oxide signalling, suppresses NF-κB.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
2 mg	2mg + 0.4 mL	5 mg/mL	1 unit = 0.05 mg
5 mg	5mg + 1 mL	5 mg/mL	1 unit = 0.05 mg
10 mg	10mg + 2 mL	5 mg/mL	1 unit = 0.05 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-2	Daily	**250 mcg**	5u	Loading. Single injection near injury site; rotate.
Weeks 3-4	Daily	**375 mcg**	7.5u	Step-up. Half-unit increment optional - many users go directly to 10u/500 mcg.
Weeks 5-10	Daily	**500 mcg**	10u	Working dose. Matches Wolverine Blend BPC-157 component.
Aggressive	Twice daily (AM + PM split)	**500 mcg × 2**	10u	Community-aggressive: 1 mg/day total split BID (researchdosing, peptidesciences). Used for acute soft-tissue injury, severe gut symptoms (Crohn's/IBD flare), or post-surgical recovery. BPC-157 has a wide therapeutic window - no toxicity signal reported up to 2 mg/day. Taper back to the daily working dose once acute phase resolves.
Off	2-4 weeks	-	-	Standard washout. Re-cycle if continued tissue work needed.
Cycle length	8-10 weeks on / 2-4 weeks off (daily microdose, matches Wolverine). Aggressive (BID): 4-6 weeks split-dose / 4 weeks off, then back to daily working dose.			
Timing	Daily, anytime — food doesn't affect; consistency matters			

Handling & Stacking

Injection site	SubQ near injury site. Rotate. Do NOT inject into joint.
Storage	Lyophilised: -4°F (years stable). Reconstituted: 36-46°F, up to 8 weeks with verified product and continuous cold chain. Discard immediately if cloudy, discoloured, or particulates.
Side effects	Usually minimal. Occasional lightheadedness or injection-site irritation.
What to expect	Pain reduction at injury site typically within days. Tissue repair progress noticeable at 3-4 weeks. Gut symptoms often improve within 1-2 weeks.
Supply math	**2mg** → 15 vials | **5mg** → 6 vials | **10mg** → 3 vials *(10-wk @ 500 mcg × 7/wk)*
Pairs well with	**TB-500** (**Wolverine stack** - synergistic soft-tissue repair); **Ipamorelin** / **CJC No DAC** (GH support during healing); **GHK-Cu** (skin/wound healing); **Tesamorelin** (GHRH support)
Avoid	NSAIDs (antagonise prostaglandin pathway). High-dose vitamin C same session (oxidation). Active malignancy. **GLOW** / **KLOW Blend** (already contains **BPC-157** - would double-dose)
References	peptidefox · researchdosing · peptidedosages · PubMed · thepeptidereport · peptides.id

Safety & Patient Education

Contraindications	• Active malignancy, pregnancy, lactation
Warnings	• No published human RCTs • FDA Category 2 (2023) — compounding from bulk substances restricted
Drug interactions	• NSAIDs: rodent data suggests blunted BPC-157 effect — clinical relevance uncertain (no human trials) • Anticoagulants (warfarin, DOACs): no human data; preclinical hints at coagulation effects — monitor INR/anti-Xa if combining
Labs	• Compound-specific: hsCRP (baseline + end)
Stop criteria	• New mass or unexpected tissue growth at any injection site • Persistent injection-site reactions >7 days (redness, induration, abscess) • New unexplained GI symptoms
Patient education	• Research compound, FDA Category 2 (2023) — compounding restricted in the US. Request COA for each lot • All human use is off-label — no published RCTs. Doses extrapolated from rodent studies and community experience • Local injection near target tissue for orthopedic use; SubQ near abdomen for systemic gut/inflammatory use. Rotate sites • Cycle: 4-6 wks on / 2-4 wks off. Continuous dosing isn't supported and may blunt response • Stop and contact clinician for new lump, persistent injection-site reaction >7 days, or GI bleeding • Do not use during active cancer or while pregnant/lactating — angiogenic and growth-promoting

BR226

BLENDS AND MIXED PROTOCOLS

In plain English	*Relaxation PM — pre-bed sleep stack: GABA + melatonin + arginine + glutamine in one shot.*
What it does	Sleep / relaxation-stack injection. GABA: inhibitory neurotransmitter; injectable bypass of poor BBB penetration is debated.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 ml	Solution no reconstitution needed	GABA 10 mg/mL Melatonin 0.1 mg/mL Arginine 100 mg/mL Glutamine 2.5 mg/mL	1 unit = 0.10 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Pre-bed	Nightly	**1 mL**	100u	Inject 30-60 min before sleep. Full vial = 10 nights.
Off	1-2 weeks	**-**	-	Optional break after 8 weeks continuous to avoid melatonin receptor downregulation.
Cycle length	4-8 weeks on / 1-2 weeks off, or PRN. Avoid daily continuous use beyond 8 weeks (melatonin receptor downregulation).			
Timing	Daily SubQ, anytime — see notes for compound-specific timing			

Handling & Stacking

Injection site	SubQ abdomen or outer thigh.
Storage	Solution: refrigerate 36-46°F. Use within 28 days. Protect from light (melatonin photodegrades).
Side effects	Vivid dreams (melatonin), AM grogginess if dosed too close to wake. Rare flushing (arginine vasodilation).
What to expect	Drowsiness within 20-30 min. Deeper subjective sleep over 1-2 weeks of consistent pre-bed use.
Pairs well with	**DSIP** (additive sleep-architecture support); **GHK-Cu** (overnight repair); **MOTS-c** off-day stacking
Avoid	Stimulants pre-bed. Daytime alertness windows (causes drowsiness).
References	PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Patient education	• Aseptic technique: alcohol swab the vial septum and the injection site, use a single-use sterile needle for every injection, never share vials or needles between people • Sharps disposal in an FDA-cleared sharps container, never household trash • Cold chain: if a refrigerated peptide reaches room temperature for more than the labeled stability window, discard it • Mark the date of first puncture on every multi-dose vial • Inspect every vial before injection: clear solution (unless labeled otherwise), no particulates, no cloudiness, no color change • Rotate injection sites to avoid lipoatrophy, induration, and infection • Carry a record of what you're injecting if you need emergency care

CJC-1295 (NO DAC)

GROWTH HORMONE AXIS

In plain English	*A short-acting version. Each shot gives you about a 30-minute GH pulse. Usually paired with Ipamorelin.*
What it does	GHRH analog producing natural pulsatile GH release when injected pre-bed. Short half-life = clean GH pulse. Binds GHRH receptors on pituitary somatotrophs.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 2.5 mL	2 mg/mL	1 unit = 0.02 mg
10 mg	10mg + 3 mL	3.33 mg/mL	1 unit = 0.03 mg

Protocol

Weeks	Frequency	Dose	Units 2 mg/mL	Units 3.33 mg/mL	Notes
Weeks 1-2	Nightly 5×/week	**200 mcg**	10u	6.0u	Start low and assess how your body responds before increasing.
Weeks 3-12	Nightly 5×/week	**300 mcg**	15u	9.0u	Standard effective dose for most users.
Off	4 weeks	**-**	-	-	Off period - let your pituitary and receptors recover before the next cycle.
Cycle length	8-12 weeks on / 4 weeks off. Max: 12 weeks on / 4-8 weeks off. 5 nights/week (M-F) preferred over daily.				
Timing	Fasted AM or fasted pre-bed. No food 60-90 min before; no food 60 min after — food-driven insulin blunts the GH pulse				

Handling & Stacking

Injection site	Abdomen. Can combine with Ipamorelin in same syringe.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, up to 6 weeks.
Side effects	Water retention, tingling, mild fatigue.
What to expect	Improved sleep quality within 1-2 weeks. Recovery improvement at 4-6 weeks. Body composition changes at 8-12 weeks.
Supply math	**5mg** → 4 vials | **10mg** → 2 vials *(12-wk @ 300 mcg × 5/wk)*
Pairs well with	**Ipamorelin** (canonical GHRH + GHRP synergy); GHRP-2 / GHRP-6 / Hexarelin (GH pulse amplification); **Wolverine Blend** (**BPC-157 + TB-500**) (recovery stack); **IGF-1 LR3** (downstream IGF amplification)
Avoid	Long-acting glucocorticoids (suppress GH axis). Active malignancy (GH may promote growth)
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox · peptides.id

Safety & Patient Education

Contraindications	• Active malignancy of any kind (GH/IGF-1 can promote tumor growth) • Severe diabetic retinopathy • Acute critical illness (per somatropin label — applies to GH-axis stimulators) • Pregnancy and lactation
Warnings	• Insulin resistance and worsening glycemic control with sustained use • Fluid retention, peripheral edema • Carpal tunnel syndrome with extended use • Joint and muscle pain • IGF-1 elevation: theoretical malignancy promotion if dosed beyond physiological IGF-1 ranges • Combined GHRH+GHRP use requires off cycles to prevent receptor desensitization
Drug interactions	• Glucocorticoids: blunt GH response — minimize or time away from injection • Insulin / antidiabetics: GH worsens insulin resistance — expect dose adjustments + tighter monitoring • Levothyroxine: GH lowers free T4 — recheck TSH/fT4 quarterly • Oral estrogens: blunt GH-stimulated IGF-1 — transdermal doesn't • CYP3A4 narrow-TI substrates (cyclosporine, tacrolimus, sirolimus): recheck troughs after starting
Labs	• Compound-specific: IGF-1 (baseline + Week 8 + end), Lipid panel (Week 0/12) • IGF-1 at baseline, mid-cycle, and end of cycle. Target ≤ULN for age • Fasting glucose and HbA1c at baseline and end of cycle • Mammogram and prostate screening per age-appropriate guidelines before starting
Stop criteria	• Fasting glucose rises >20 mg/dL above baseline • IGF-1 rises above the upper limit of normal for age • New persistent joint pain or carpal tunnel symptoms • New skin changes or moles enlarging • Vision changes
Patient education	• GH-axis stimulation is not appropriate during active malignancy or for fertility • Most GH pulses occur during deep sleep — dose pre-bed unless instructed otherwise • Food within 90 min of injection blunts GH response (carbs/fats most) — fasted is standard • Report new lumps, persistent headaches, or vision changes immediately

CJC-1295 (NO DAC) + IPAMORELIN

GROWTH HORMONE AXIS

In plain English	*A pre-mixed combination of two different growth hormone stimulators that work via separate pathways, producing a synergistic pulse that's stronger than either alone - without increasing cortisol or appetite.*
What it does	Pre-mixed GHRH + GHRP stack - amplified, synergistic GH pulse from two different receptor pathways in one injection. CJC-1295 (No DAC) stimulates GHRH receptors. Ipamorelin activates GHS-R1a ghrelin receptors.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 3 mL	3.33 mg/mL	1 unit = 0.03 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-2	Nightly pre-bed	**200 mcg**	6.0u	Start low and assess how your body responds before increasing.
Weeks 3-12	Nightly pre-bed	**400 mcg**	12.0u	Standard effective dose for most users.
Off	4 weeks	**-**	-	Off period - let your pituitary and receptors recover before the next cycle.
Cycle length	12 weeks on / 4 weeks off. Max 16 weeks on / 4-8 weeks off; receptor desensitization beyond 16 weeks continuous.			
Timing	Daily SubQ, anytime — see notes for compound-specific timing			

Handling & Stacking

Injection site	Abdomen. Single injection delivers both.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 4 weeks.
Side effects	Water retention, mild fatigue, tingling.
What to expect	Improved sleep depth within 1-2 weeks. Recovery and energy improvement at 4-6 weeks. Body composition changes at 8-12 weeks.
Supply math	**10mg** → 4 vials *(12-wk @ 400 mcg × 7/wk)*
Pairs well with	**BPC-157** / **TB-500** (tissue repair on GH backbone); **IGF-1 LR3** (downstream amplification)
Avoid	Active malignancy. Type 2 diabetes without monitoring (GH worsens insulin resistance)
References	peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Active malignancy of any kind (GH/IGF-1 can promote tumor growth) • Severe diabetic retinopathy • Acute critical illness (per somatropin label — applies to GH-axis stimulators) • Pregnancy and lactation
Warnings	• Insulin resistance and worsening glycemic control with sustained use • Fluid retention, peripheral edema • Carpal tunnel syndrome with extended use • Joint and muscle pain • IGF-1 elevation: theoretical malignancy promotion if dosed beyond physiological IGF-1 ranges • Combined GHRH+GHRP use requires off cycles to prevent receptor desensitization
Drug interactions	• Glucocorticoids: blunt GH response — minimize or time away from injection • Insulin / antidiabetics: GH worsens insulin resistance — expect dose adjustments + tighter monitoring • Levothyroxine: GH lowers free T4 — recheck TSH/fT4 quarterly • Oral estrogens: blunt GH-stimulated IGF-1 — transdermal doesn't • CYP3A4 narrow-TI substrates (cyclosporine, tacrolimus, sirolimus): recheck troughs after starting
Labs	• Compound-specific: IGF-1 (baseline + Week 8 + end), Lipid panel (Week 0/12) • IGF-1 at baseline, mid-cycle, and end of cycle. Target ≤ULN for age • Fasting glucose and HbA1c at baseline and end of cycle • Mammogram and prostate screening per age-appropriate guidelines before starting
Stop criteria	• Fasting glucose rises >20 mg/dL above baseline • IGF-1 rises above the upper limit of normal for age • New persistent joint pain or carpal tunnel symptoms • New skin changes or moles enlarging • Vision changes
Patient education	• GH-axis stimulation is not appropriate during active malignancy or for fertility • Most GH pulses occur during deep sleep — dose pre-bed unless instructed otherwise • Food within 90 min of injection blunts GH response (carbs/fats most) — fasted is standard • Report new lumps, persistent headaches, or vision changes immediately

DSIP

COGNITIVE AND NEUROPROTECTIVE

In plain English	*A natural sleep peptide. Pushes you toward deeper sleep and tones down nighttime stress hormones. Inject before bed.*
What it does	Delta Sleep-Inducing Peptide - promotes slow-wave (delta) sleep, reduces stress hormones, and regulates circadian rhythm. Modulates CRH-. May regulate pineal

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
2 mg	2mg + 1 mL	2 mg/mL	1 unit = 0.02 mg
5 mg	5mg + 2.5 mL	2 mg/mL	1 unit = 0.02 mg
10 mg	10mg + 3 mL	3.33 mg/mL	1 unit = 0.03 mg

Protocol

Weeks	Frequency	Dose	Units 2 mg/mL	Units 3.33 mg/mL	Notes
Weeks 1-2	Nightly	**200 mcg**	10u	6.0u	Start low and assess how your body responds before increasing.
Weeks 3-8	Nightly	**200-500 mcg**	10-25u	6.0-15.0u	Effective range.
Off	2-4 weeks	**-**	-	-	Essential break between cycles. Skipping leads to tolerance and diminishing returns.
Cycle length	4-8 weeks on, 2-4 weeks off.				
Timing	Pre-bed nightly — 15-30 min before sleep				

Handling & Stacking

Injection site	SubQ abdomen.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 2 weeks.
Side effects	Sedation (expected), headache (occasional), vivid dreams.
What to expect	Deeper, more restorative sleep typically within the first few doses. Reduced overnight cortisol. Vivid dreams in some users.
Supply math	**2mg** → 12 vials | **5mg** → 5 vials | **10mg** → 3 vials *(8-wk @ 500 mcg × 7/wk)*
Pairs well with	Oral melatonin (sleep architecture); Glycine (deep-sleep amplification); **GHK-Cu** (recovery during sleep)
Avoid	Active or history of cancer (theoretical GH-related concern). Pregnancy / nursing
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox · peptides.id

Safety & Patient Education

Contraindications	• Pregnancy, lactation, active or history of cancer (theoretical GH-related concern)
Warnings	• Limited human data
Drug interactions	• CNS depressants (benzodiazepines, Z-drugs, barbiturates, opioids, alcohol): additive sedation — avoid same-night combo or reduce dose • Other sleep peptides (oral melatonin, Epithalon): redundant rather than additive — pick one per cycle • Sedating antidepressants (mirtazapine, trazodone, doxepin, amitriptyline): additive sedation — coordinate timing • Stimulants (caffeine, ADHD meds): mechanistically opposed — not harmful but blunts effect • Antihypertensives: theoretical additive nocturnal BP drop — monitor if combining at high doses
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Stop criteria	• Excessive sedation, vivid dreams disturbing sleep, persistent daytime grogginess
Patient education	• Aseptic technique: alcohol swab the vial septum and the injection site, use a single-use sterile needle for every injection, never share vials or needles between people • Sharps disposal in an FDA-cleared sharps container, never household trash • Cold chain: if a refrigerated peptide reaches room temperature for more than the labeled stability window, discard it • Mark the date of first puncture on every multi-dose vial • Inspect every vial before injection: clear solution (unless labeled otherwise), no particulates, no cloudiness, no color change • Rotate injection sites to avoid lipoatrophy, induration, and infection • Carry a record of what you're injecting if you need emergency care

EPITHALON

LONGEVITY AND BIOREGULATORS

In plain English	*Russian peptide that activates telomerase — the enzyme that keeps the ends of your chromosomes from fraying. The most-studied of the longevity peptides.*
What it does	Research-phase peptide; no published human RCTs. Telomerase activation reported in rodent studies. Stimulates telomerase (hTERT) expression. Normalises pineal Developed by Vladimir Khavinson.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 0.50 mL	20 mg/mL	1 unit = 0.20 mg
50 mg	50mg + 2.5 mL	20 mg/mL	1 unit = 0.20 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Days 1-10	Daily or twice daily	**5-10 mg**	25-50u	10 mg/day × 10 days standard.
Days 1-20	Daily	**5 mg**	25u	Extended course.
Repeat	Seasonal	**2-3×/year**	-	4-6 months apart.
Cycle length	10-20 day courses, 2-3× per year.			
Timing	10-day course, daily SubQ pre-bed — repeat 2×/year			

Handling & Stacking

Injection site	SubQ abdomen or outer thigh.
Storage	Lyophilised: -4°F (24 months). Reconstituted: 36-46°F, use within 20 days.
Side effects	Generally very well tolerated. Rare: injection-site irritation.
What to expect	Improved sleep and circadian normalisation first (1-2 weeks). Gradual energy and immune improvement over 2-3 annual courses. Benefits compound over years.
Supply math	**10mg** → 27 vials | **50mg** → 6 vials *(4-wk @ 10 mg × 7/wk)*
Pairs well with	**Thymosin Alpha-1** (immune amplification); **DSIP** (sleep architecture, complementary circadian effect)
Avoid	Active malignancy. Pregnancy / nursing
References	perfectb · researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Pregnancy and lactation (no safety data) • Active malignancy (theoretical concerns; no human RCT data on cancer risk) • Children and adolescents (no pediatric data outside Russian clinical use)
Warnings	• Most clinical evidence is Russian/CIS — much doesn't meet ICH-GCP standards • No FDA approval; not regulated as drugs in the US • Limited human safety data outside the Khavinson research group • Long-term safety beyond 5 years of cyclic use not established • Most-studied of the Khavinson peptides but still no Western RCTs
Labs	• Compound-specific: Salivary melatonin at 9 PM (baseline + after 20-day course) • No specific markers. Baseline CBC + CMP; target-tissue labs (salivary melatonin for Epithalon)
Patient education	• Bioregulators — research use only in most jurisdictions • Russian use approved; US/EU off-label or unapproved • Quality varies — request COA for each lot • 10-day courses, 2-3×/year — continuous dosing isn't supported

GHK-CU

TISSUE REPAIR AND RECOVERY

In plain English	*A peptide that carries copper into your cells and flips on the repair genes for skin, hair, and connective tissue. Your body makes it, but levels crash about 90% by age 60.*
What it does	Copper-binding tripeptide - activates 4,000+ repair genes. Plasma levels decline 90% between age 20 and 60. Activates genes for collagen/elastin synthesis, angiogenesis, and anti-inflammation.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
50 mg	50mg + 5 mL	10 mg/mL	1 unit = 0.10 mg
100 mg	100mg + 10 mL	10 mg/mL	1 unit = 0.10 mg (reconstitute into 10 mL BAC water vial)

Protocol

Weeks	Frequency	Dose	Units	Notes
Week 1	Daily	**1 mg**	10u	Loading. Establish tolerance; watch for injection-site sting or redness.
Week 2	Daily	**1.5 mg**	15u	Ramp. Smooth step-up from loading dose.
Weeks 3-8	Daily	**2 mg**	20u	Working dose. Full effect builds over the next 4-6 weeks.
Aggressive	1-2× daily AM	**3-4 mg**	30-40u	Bump here ONLY if: (1) you've completed at least 2 weeks at 2 mg with zero injection-site sting, headache, or histamine flare, AND (2) you have a specific high-demand goal - post-surgical recovery, active wound healing, or a stalled hair/skin response after 6+ weeks at 2 mg. Start at 3 mg (e.g., 2 mg AM + 1 mg PM to leverage GHK-Cu's short half-life). Push to 4 mg only if 3 mg is producing visible results and you want to test further - PeptideDeck and SeekPeptides note no published benefit above 3 mg/day; cost roughly doubles.
Off	4 weeks	**-**	-	Mandatory copper-accumulation washout. Do not skip - especially after an Aggressive phase.
Cycle length	8 weeks on / 4 weeks off. Stays within copper-accumulation limits while giving 6 weeks at working dose.			
Timing	Daily SubQ anytime, or topical twice daily			

Handling & Stacking

Injection site	SubQ adjacent to target. Systemic: abdomen or thigh.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, light-protected (copper oxidizes), 6 weeks. Use amber vial or foil wrap.
Side effects	Rare injection-site irritation. Copper dysregulation with very extended continuous use.
What to expect	Skin quality improvement at 6-8 weeks. Hair growth improvement at 8-12 weeks. Localised tissue improvement adjacent to injection sites.
Supply math	**50mg** → 2 vials | **100mg** → 2 vials *(8-wk @ 2 mg × 7/wk)* *Use 50 mg vials at the 2 mg working dose — reconstituted shelf life is ~28 days; the 100 mg vial only depletes inside that window at the Aggressive 3-4 mg/day phase.*
Pairs well with	**BPC-157** (tissue repair stack); **TB-500** (collagen synthesis); **CJC-1295 No DAC + Ipamorelin** (GH environment for skin); **Tesamorelin** (GHRH adjunct)
Avoid	High-dose vitamin C same session (copper oxidation). Extended continuous use (mandatory 30-day off). **GLOW** / **KLOW Blend** (already contains **GHK-Cu** - would double-dose copper)
References	peptidedeck · seekpeptides · peptidedosingprotocols · researchdosing · peptidedosages · PubMed

Safety & Patient Education

Contraindications	• Wilson's disease, copper dysregulation, copper allergy
Warnings	• Extended continuous use can cause copper accumulation; the document's mandatory 30-day off period is critical
Drug interactions	• High-dose vitamin C (oral or IV): theoretical copper-ascorbate redox — separate timing • Copper chelators (penicillamine, trientine, tetrathiomolybdate): directly antagonistic • High-dose zinc (>50 mg/day chronic): reduces copper absorption — long-term combo blunts effect • High-dose iron: competes with copper for transport — reduces uptake • Topical retinoids / AHAs: irritation if same area — separate sessions
Labs	• Compound-specific: hsCRP (baseline + end) • Serum copper and ceruloplasmin if cycles are extended
Stop criteria	• Skin discoloration, neurological symptoms (copper toxicity)
Patient education	• Aseptic technique: alcohol swab the vial septum and the injection site, use a single-use sterile needle for every injection, never share vials or needles between people • Sharps disposal in an FDA-cleared sharps container, never household trash • Cold chain: if a refrigerated peptide reaches room temperature for more than the labeled stability window, discard it • Mark the date of first puncture on every multi-dose vial • Inspect every vial before injection: clear solution (unless labeled otherwise), no particulates, no cloudiness, no color change • Rotate injection sites to avoid lipoatrophy, induration, and infection • Carry a record of what you're injecting if you need emergency care

GLOW BLEND

BLENDS AND MIXED PROTOCOLS

In plain English	*Wolverine (BPC + TB) plus GHK-Cu. Adds skin, hair, and connective-tissue support to the basic healing stack.*
What it does	Three-peptide regenerative blend: + TB-500 (systemic repair) + BPC-157 (healing). Pre-mixed convenience. 000+ repair genes. TB-500 upregulates thymosin β4 for systemic repair. BPC-157 drives local angiogenesis.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
70 mg	70mg + 5 mL	GHK-Cu 10 mg/mL BPC-157 2 mg/mL TB-500 2 mg/mL	1 unit = 0.10 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Week 1	Daily	**1 mg**	10u	Loading. 1 mg GHK-Cu + 0.2 mg BPC-157 + 0.2 mg TB-500 per dose.
Week 2	Daily	**1.5 mg**	15u	Ramp. 1.5 mg GHK-Cu + 0.3 mg BPC + 0.3 mg TB per dose.
Weeks 3-8	Daily	**2 mg**	20u	Working dose. 2 mg GHK-Cu + 0.4 mg BPC + 0.4 mg TB per dose.
Aggressive	Daily (AM + PM split)	**3-4 mg**	30-40u	Bump here ONLY if 2+ weeks at 2 mg with zero injection-site issues AND a specific high-demand goal. Start at 3 mg (= 0.6 mg each of BPC/TB). Push to 4 mg only if 3 mg is producing visible results and you want to test further - no published benefit above 3 mg/day; cost roughly doubles.
Off	4 weeks	**-**	-	Mandatory copper-accumulation washout. Do not skip - especially after an Aggressive phase.
Cycle length	8 weeks on / 4 weeks off - matches GHK-Cu copper-accumulation limits.			
Timing	Daily SubQ, anytime — same as the Wolverine base			

Handling & Stacking

Injection site	SubQ near target tissue. Rotate.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, protected from light, 6 weeks (rate-limited by GHK-Cu component).
Side effects	Usually minimal. Injection-site irritation occasionally.
What to expect	Broad tissue repair, skin quality improvement, and anti-inflammatory effects build over the 8-week cycle. 4-week washout follows.
Supply math	**70mg** → 2 vials *(8-wk @ 2 mg × 7/wk)* *70 mg vial covers ~half of an 8-week cycle at 2 mg/day; reconstituted shelf life ~28 days. Aggressive 3-4 mg/day depletes the vial faster but stays inside the discard window.*
Pairs well with	**CJC-1295 No DAC + Ipamorelin** (systemic GH + local repair); **Thymosin Alpha-1** (immune support); **Epithalon** (longevity / collagen complement)
Avoid	NSAIDs (antagonise prostaglandin pathway, blunting the blend's repair signalling). High-dose vitamin C same session (copper oxidation)
References	peptidefox · researchdosing · peptidedosages · PubMed · thepeptidereport · peptides.id

Safety & Patient Education

Contraindications	• Active malignancy (TB-500 promotes angiogenesis; BPC-157 promotes tissue growth) • Wilson's disease, copper dysregulation, copper allergy (GHK-Cu component) • Active bleeding or recent hemorrhagic stroke (TB-500 vascular effects) • Pregnancy, lactation
Warnings	• Combines three compounds with no human RCT data on the blend itself. • Quality varies by compounder; request COA listing each component's purity. • GHK-Cu: extended continuous use can cause copper accumulation — the 6-week off period is critical. • BPC-157: FDA Category 2 in 2023; compounding from bulk substances is restricted. • TB-500: no published human efficacy trials; theoretical malignancy risk via angiogenesis.
Drug interactions	• NSAIDs: rodent data suggests blunted BPC-157 effect; clinical relevance uncertain (no human data) • High-dose vitamin C same session: theoretical copper-ascorbate redox interaction with GHK-Cu • Anticoagulants (warfarin, DOACs): caution with TB-500 vascular effects
Labs	• Compound-specific: hsCRP (baseline + end) • Serum copper and ceruloplasmin if cycles extend beyond 6 weeks (GHK-Cu accumulation)
Stop criteria	• New mass or unexpected tissue growth at any injection site • Skin discoloration, neurological symptoms (copper toxicity from GHK-Cu) • Persistent injection-site reactions beyond 7 days
Patient education	• Three-peptide blend: any single-component contraindication applies to the whole blend. • Cycle discipline matters more than dose: 6 weeks on / 6 weeks off is the canonical pattern. • Inject SubQ near target tissue. Rotate sites to reduce GHK-Cu sting.

GLUTATHIONE

SUPPLEMENTS AND COFACTORS

In plain English	*Your body's master antioxidant. Injected for liver support, brighter skin, and bouncing back from oxidative stress. Mix it fresh — it breaks down fast.*
What it does	Master antioxidant tripeptide central to oxidative stress defence, liver detox, and immune function. Directly neutralises ROS, regenerates vitamins C and E, supports liver Phase II detox, and maintains mitochondrial integrity.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
600 mg	600mg + 1.2 mL	500 mg/mL	1 unit = 5.00 mg
1500 mg	1500mg + 3 mL	500 mg/mL	1 unit = 5.00 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-2	Mon-Fri upper glute	**100 mg**	20u	Tolerance check at 100 mg. 5×/week = 500 mg/week. Watch for garlic taste, flushing, or injection-site sting.
Weeks 3-8	Mon/Wed/Fri upper glute	**200 mg**	40u	Working dose ceiling (PeptideDosages canonical). 3×/week = 600 mg/week. Matches Empower/AgelessRx clinic 200 mg dose. Hold at 100 mg if 200 mg poorly tolerated.
Maintenance	1-2×/week	**200 mg**	40u	Ongoing antioxidant support. Drop to once-weekly after the 8-week loading + working phase if benefit holds.
Off	2-4 weeks	**-**	-	Essential break. Skipping leads to tolerance and diminishing returns.
Cycle length	4 weeks on / 4 weeks off. Max: 12 weeks on / 4-8 weeks off. Hepatic methylation cycling supports best results with breaks.			
Timing	Daily SubQ, reconstitute fresh — oxidizes within hours			

Handling & Stacking

Injection site	Upper outer glute preferred. Abdomen acceptable.
Storage	Lyophilised: -4°F (12-24 months). Reconstitute fresh before each injection - oxidises within hours.
Side effects	Upper glute sting (brief). Flushing or garlic taste within 30 min. Reconstitute fresh only.
What to expect	Energy improvement and reduced oxidative fatigue within 2-4 weeks. Skin brightening reported by many users. Must reconstitute fresh - oxidises within hours of reconstitution.
Supply math	**600mg** → 8 vials | **1500mg** → 3 vials *(8-wk @ 200 mg × 3/wk)*
Pairs well with	**NAD+** (cellular redox stack); **GHK-Cu** (additive antioxidant); Vitamin C (recycle **glutathione**); **BPC-157** (gut + systemic recovery); **MOTS-c** (mitochondrial)
Avoid	Sulfa allergy (theoretical). Concurrent chemotherapy (may interfere - physician supervision)
References	realpeptides · mypeptidematch · researchdosing · peptidedosages · DailyMed · PubMed

Safety & Patient Education

Contraindications	• Sulfa allergy (theoretical, reaction-prone individuals), concurrent chemotherapy without oncology supervision
Warnings	• Must be reconstituted fresh; oxidizes within hours. • FDA has flagged compounded IV glutathione for skin-lightening use as an unproven indication with serious adverse event reports.
Drug interactions	• Chemotherapy (cisplatin, doxorubicin, cyclophosphamide and others): glutathione may reduce both toxicity AND efficacy — do NOT combine outside an oncologist-supervised protocol • Acetaminophen overdose treatment (N-acetylcysteine pathway): glutathione is the protected metabolite — supplemental IV/SubQ glutathione may augment NAC therapy but is not a substitute • Warfarin: theoretical mild potentiation via reduced oxidative stress on coagulation factors; monitor INR if combining chronically • Sulfa drugs (sulfonamide antibiotics, sulfasalazine, some diuretics): increased reaction risk in sulfa-allergic patients given the sulfhydryl group • Methylating agents and SAMe: complementary methylation/redox support; theoretical synergy, no human interaction data
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Stop criteria	• Rash, hives, or sulfhydryl-type allergic reaction; severe injection-site irritation
Patient education	• Reconstitute immediately before use; discard unused — glutathione oxidizes within hours and is no longer active. • Do not use during active cancer treatment without oncology coordination. • Skin-lightening use is unproven and the FDA has issued safety warnings about compounded IV products marketed for that purpose.

HCG

REPRODUCTIVE AND HORMONAL

In plain English	*A pregnancy hormone that looks like LH to your body. Men on TRT use it to keep their testicles working and stay fertile.*
What it does	Human Chorionic Gonadotropin - mimics LH to maintain testicular function and testosterone during suppressive cycles. Binds LH/CG receptors on Leydig cells to stimulate testosterone production.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
1000 iu	1000IU + 0.5 mL	2000 IU/mL	U-100 syringe
2000 iu	2000IU + 1 mL	2000 IU/mL	U-100 syringe
5000 iu	5000IU + 2.5 mL	2000 IU/mL	U-100 syringe
10000 iu	10000IU + 5 mL	2000 IU/mL	U-100 syringe

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	Mon/Wed/Fri	**250 IU**	12.5u	LH-mimetic - maintains testosterone.
Higher	Mon/Wed/Fri	**500 IU**	25u	At this dose, estrogen (E2) elevation is more likely. Have an AI available and test at Week 4.
Fertility	Daily or M/W/F	**500-1,000 IU**	25-50u	Full fertility support dose. This protocol takes 70+ days to show results in sperm counts.
Cycle length	TRT-adjunct: ongoing while on TRT (no scheduled off-cycle). Fertility: 70+ day cycle (one full spermatogenesis cycle). Single-dose ovulation trigger: one IM injection per IVF cycle (per FDA Pregnyl / Novarel labels).			
Timing	2-3×/week SubQ, anytime — same days each week			

Handling & Stacking

Injection site	SubQ abdomen. Rotate.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 60 days.
Side effects	Water retention, mood swings, estrogen elevation.
What to expect	Testicular volume maintenance and testosterone support within days. Monitor estrogen - HCG raises both T and E2.
Supply math	**1000iu** → 14 vials | **2000iu** → 7 vials | **5000iu** → 3 vials | **10000iu** → 2 vials *(10-wk @ 500 iu × 3/wk)*
Pairs well with	**CJC-1295 No DAC + Ipamorelin** (systemic anabolic environment); **Kisspeptin** (HPG-axis stimulation)
Avoid	Hormone-sensitive cancer (prostate, breast). Active polycythaemia. Untreated thrombophilia (estradiol elevation)
References	rxlist · clevelandclinic · defymedical · researchdosing · peptidedosages · DailyMed

Safety & Patient Education

Contraindications	• Hormone-sensitive malignancy (prostate cancer outside intentional ADT, breast cancer, ovarian cancer) • Active thromboembolic disease • Untreated thrombophilia • Pregnancy (except specific fertility indications under specialist supervision) • Polycythemia vera
Warnings	• Estradiol elevation common; aromatase inhibitor may be needed • Thromboembolic risk with elevated estradiol • Ovarian hyperstimulation syndrome (OHSS) in females on HCG: can be life-threatening • Mood changes, gynecomastia in males • HCG raises both T and E2; estradiol monitoring is mandatory • OHSS in females, thromboembolism (estradiol elevation), precocious puberty in males with cryptorchidism
Drug interactions	• AIs (anastrozole, letrozole): co-administered to manage E2 — over-suppression worsens lipids and bone • SERMs (clomiphene, tamoxifen, enclomiphene): redundant overlap with HCG via central HPG drive • Exogenous testosterone: shuts down LH/FSH — HCG efficacy reduced unless TRT paused or microdosed • Anticoagulants (warfarin, DOACs): elevated E2 raises thromboembolic risk — coordinate if chronic • Insulin / antidiabetics: HCG may shift glucose at high doses — monitor diabetic patients
Labs	• Compound-specific: Estradiol E2 (baseline + Week 4-6), Total & Free Testosterone (baseline + end) • Total + free testosterone, LH, FSH, sensitive E2 at baseline, mid-cycle, end • SHBG and prolactin at baseline • CBC + hematocrit (polycythemia surveillance) • Lipid panel; PSA in men >40 at baseline • Semen analysis baseline + quarterly for fertility protocols • D-dimer if clotting symptoms • Estradiol, hematocrit
Stop criteria	• Hematocrit >54% — donate blood or hold therapy • Estradiol persistently above target despite AI • OHSS symptoms (abdominal distension, weight gain, dyspnea) • Thrombosis symptoms (calf swelling, chest pain, dyspnea, neuro) • New breast tenderness or growth (gynecomastia workup)
Patient education	• Watch for clotting symptoms: leg swelling, calf pain, chest pain, sudden shortness of breath • Estrogen side effects (gynecomastia, mood) manageable with AI under physician guidance • Fertility protocols: 70+ days before semen analysis changes • HCG raises estrogen significantly • Recon math here reflects community TRT-adjunct / PCT use (multi-dose SubQ). FDA labels (Pregnyl, Novarel, Ovidrel) describe single-use IM only — off-label is community-standard but unlabeled.

HGH (SOMATROPIN)

GROWTH HORMONE AXIS

In plain English	*Actual growth hormone in a vial. Sold as Norditropin, Genotropin, and others. The real thing — no triggering involved.*
What it does	Recombinant human growth hormone - augments GH for body composition, recovery, anti-aging, and metabolic benefits. Binds GHR receptor activating JAK2/STAT5 for IGF-1 production, protein synthesis, fat lipolysis, and cellular repair.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
6 iu	6IU (2mg) + 0.4 mL	5 mg/mL	1 unit = 0.05 mg
8 iu	8IU (2.67mg) + 0.53 mL	5 mg/mL	1 unit = 0.05 mg
10 iu	10IU (3.33mg) + 0.67 mL	5 mg/mL	1 unit = 0.05 mg
12 iu	12IU (4mg) + 0.8 mL	5 mg/mL	1 unit = 0.05 mg
15 iu	15IU (5mg) + 1 mL	5 mg/mL	1 unit = 0.05 mg
24 iu	24IU (8mg) + 1.6 mL	5 mg/mL	1 unit = 0.05 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Anti-aging	Daily pre-bed	**1-2 IU/day**	7-13u	Conservative anti-aging range. Gentle on insulin sensitivity. Good long-term starting point.
Body comp	Daily or split	**2-4 IU/day**	13-27u	The most commonly used body-composition dose. Expect water retention in weeks 1-4 as tissues rehydrate - this is normal.
Advanced	Split AM+PM	**4-6+ IU/day**	27-40u	Higher performance dose. At this level, insulin resistance and carpal tunnel become relevant - regular blood work is important.
Off	Equal to on-time	**-**	-	Match your off-time to your on-time. 3 months on = 3 months off. This preserves long-term GH receptor sensitivity.
Cycle length	Ongoing per FDA somatropin labels (Norditropin, Genotropin, Humatrope, Saizen, Omnitrope, Zomacton) for adult and pediatric GHD. No scheduled off-cycle in FDA-approved use. Off-label adult anti-aging use sometimes cycles 3-6 months on / 3-6 months off - that pattern is not FDA-labeled.			
Timing	Pre-bed nightly (preferred) or fasted AM. No food 60-90 min before; no food 60 min after — pulse-mimicking dosing			

Handling & Stacking

Injection site	SubQ abdomen. Never IM.
Storage	Lyophilised: 36-46°F (do NOT freeze). Reconstituted: 36-46°F, Up to 28 days with continuous refrigeration. Never freeze reconstituted.
Side effects	Water retention, joint pain, carpal tunnel syndrome, insulin resistance.
What to expect	Sleep and recovery improvement within 2-4 weeks. Body composition changes over 3-6 months. Water retention common in first 4-6 weeks. Joint comfort improves by week 6-8.
Supply math	**10iu** → 11 vials | **12iu** → 9 vials | **15iu** → 7 vials | **24iu** → 5 vials *(4-wk @ 4 iu × 7/wk)*
Pairs well with	**Ipamorelin** (additive but redundant - usually choose one); **Tesamorelin** (visceral fat); **BPC-157** / **TB-500** (joint support); **IGF-1 LR3** (cutting cycles)
Avoid	Active malignancy (FDA contraindication). Severe diabetic retinopathy. Acute critical illness (FDA black-box).
References	peptidedosages · DailyMed · PubMed · ClinicalTrials · Drugs.com

HGH (SOMATROPIN)

GROWTH HORMONE AXIS — SAFETY & PATIENT EDUCATION

Contraindications	• Active malignancy • Acute critical illness (post-surgical, ICU): increased mortality reported in trials • Severe / proliferative diabetic retinopathy • Pregnancy and lactation
Warnings	• Intracranial hypertension with somatropin (papilledema, vision changes, headache, nausea, vomiting) • Insulin resistance and new-onset T2DM • Fluid retention, edema, carpal tunnel • Pancreatitis (rare) • IGF-1 elevation above age range carries theoretical malignancy risk • Boxed warning: do not use in acutely critically ill patients (post-surgical, ICU) • Intracranial hypertension surveillance • Slipped capital femoral epiphysis in pediatrics • Scoliosis progression in pediatrics • Pancreatitis reported (rare)
Drug interactions	• Glucocorticoids: blunt GH/IGF-1 response — reassess steroid dose after starting GH • Insulin / antidiabetics: GH raises insulin needs; IGF-1 and IGF-1 LR3 cause hypoglycemia — adjust accordingly • Levothyroxine: GH lowers free T4 — recheck TSH/fT4 quarterly • Oral estrogens: blunt GH-stimulated IGF-1 — consider transdermal if response is blunted • CYP3A4 narrow-TI substrates (cyclosporine, tacrolimus, sirolimus): recheck troughs after starting • Anticonvulsants (phenytoin, carbamazepine, phenobarbital): GH may shift seizure threshold — monitor levels
Labs	• Compound-specific: IGF-1 (baseline + Week 8 + end), Fasting glucose (Week 0/4/8), Liver enzymes AST/ALT (baseline), Lipid panel (Week 0/12) • IGF-1 at baseline and quarterly during cycles • Fasting glucose, HbA1c quarterly • TSH and free T4 quarterly (GH affects thyroid axis) • Cortisol baseline (GH can mask central adrenal insufficiency) • Lipid panel • Hip/scoliosis exam in pediatrics • TSH and free T4 quarterly (GH lowers free T4)
Stop criteria	• Severe headache with vision changes (rule out intracranial hypertension) • New onset hyperglycemia • Persistent edema • New-onset hip pain in pediatrics
Patient education	• Vials must be refrigerated never frozen (somatropin) • Rotate injection sites • Report headaches with nausea/vision changes immediately • Report new joint pain • Children: report any limp or hip/knee pain immediately • Strict cold chain. Never freeze reconstituted. Never reuse. • Pen-device sharing prohibited even with a fresh needle (Norditropin, Genotropin, Humatrope, Saizen, Omnitrope, Zomacton are pen-delivered FDA-approved somatropin products; cartridge contamination from prior puncture is documented). • Report headaches with vision changes immediately.

HHB

BLENDS AND MIXED PROTOCOLS

In plain English	*A B-vitamin + choline/inositol injection blend (9 components) for hair, skin, nails, and methylation support.*
What it does	Healthy Hair, Skin, Nails Blend - injectable B-complex (9 components) + choline/inositol for keratin synthesis, methylation, and s Nine B-vitamins support keratin synthesis, collagen production, NAD+ precursor loading, and cellular energy.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 ml	Solution no reconstitution needed	Niacinamide 50 mg/mL Thiamine HCl 50 mg/mL Pantothenic 25 mg/mL Choline 10 mg/mL Inositol 10 mg/mL Niacin 5 mg/mL Biotin 0.1 mg/mL Folic 0.1 mg/mL Riboflavin 0.1 mg/mL	1 unit = 0.50 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Loading (Weeks 1-4)	2-3×/week	**0.5-1 mL**	50-100u	Loading phase establishes B-vitamin saturation. Watch for niacin flush at higher per-injection volume.
Maintenance	Weekly	**1 mL**	100u	Maintenance dose once vitamin levels are saturated. Hair/nail visible effects build 8-12 weeks.
Off	2 weeks	**-**	-	Optional break between 8-12 week maintenance blocks.
Cycle length	8-12 weeks on / 2 weeks off, or ongoing.			
Timing	3×/week IM/SubQ, anytime			

Handling & Stacking

Injection site	Upper outer glute or thigh.
Storage	Solution: refrigerate 36-46°F. Use within 28 days. Protect from light.
Side effects	Niacin flush at higher per-injection doses. Possible injection-site stinging.
What to expect	Energy and methylation support within 1-2 weeks. Hair and nail effects build over 8-12 weeks.
Pairs well with	**NAD+** (B-vitamin synergy); **Glutathione** (methylation pairing); GLP-1s (B-vitamin depletion during caloric restriction)
Avoid	B-vitamin allergy (rare).
References	peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Component allergies (B12, riboflavin, lecithin, choline, methionine, lidocaine) • Hypothyroid medication — separate by ≥4 hours • Severe hepatic impairment (choline metabolism) • Pregnancy and lactation — specific compounding required • Bleeding disorders or anticoagulants (relative) — bruising risk
Warnings	• Compounded products — composition varies; request the pharmacy's COA • B12 (cyanocobalamin) contains cobalt — allergy possible • Methionine elevates homocysteine in some individuals • L-Carnitine: TMAO conversion concern in CV disease (controversial — baseline lipids) • Injection-site reactions common with phosphatidylcholine
Drug interactions	• Warfarin: L-carnitine and choline blends can potentiate — monitor INR if combining • Levothyroxine: separate by ≥4 hours — choline/B-vitamins affect absorption • Anticoagulants / antiplatelets (DOACs, aspirin, clopidogrel): bruising risk with phosphatidylcholine • Insulin / antidiabetics: B-complex may shift glucose — recheck log if dosing changes • Methotrexate, trimethoprim, phenytoin: deplete folate / B12 — coordinate with prescriber • Lidocaine blends (LC526): avoid other caines same session — additive cardiac effects • Stimulants / phentermine: additive CV load with high-dose lipotropics — monitor BP/HR
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required • Baseline B12, folate, MMA, homocysteine (methylation markers) • TSH at baseline (B12 deficiency masks hypothyroidism) • Liver enzymes (choline metabolism) • Hematocrit (B12 raises RBC — polycythemia risk in some)
Patient education	• Fishy odor from choline/L-carnitine is normal — not allergic • Inject slowly to reduce stinging • Don't use with warfarin without physician guidance (potentiation risk) • Request exact ingredient list and concentrations from your compounder

IGF-1 (STANDARD)

GROWTH HORMONE AXIS

In plain English	*The growth signal that GH normally creates downstream. Injecting it directly skips a step. Tanks blood sugar — eat carbs with it.*
What it does	Standard insulin-like growth factor 1 - the natural form with a 12-15 min half-life. Binds IGF-1 receptor (IGF-1R) on muscle cells to activate PI3K/AKT and MAPK/ERK pathways, promoting protein synthesis, satellite c

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
1 mg	1mg + 2.5 mL	0.4 mg/mL	1 unit = 0.004 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Entry	Post-workout daily	**20-40 mcg**	5-10u	Start here. Assess hypoglycemia response before increasing.
Working	Post-workout daily	**60-80 mcg**	15-20u	Community working dose. Have fast carbs available.
Max	Post-workout daily	**100 mcg**	25u	Upper community range. Do not exceed without monitoring.
Off	4 weeks	**-**	-	Receptor sensitivity reset.
Cycle length	4-6 weeks on, 4 weeks off. Short cycles only.			
Timing	With food — fast-acting carbs available; never fasted (hypoglycemia risk)			

Handling & Stacking

Injection site	SubQ or IM directly into trained muscle for site-specific effect.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 10 days. Unstable - use promptly.
Side effects	Hypoglycemia (primary risk - carry fast carbs), injection-site edema, jaw pain, carpal tunnel with prolonged use.
What to expect	Localised anabolic response at the injection site within minutes of administration. Systemic effect is minimal due to rapid IGFBP binding. Muscle fullness and pump lasting 1-2 hours post-injection.
Supply math	**1mg** → 3 vials *(4-wk @ 80 mcg × 7/wk)*
Pairs well with	**CJC-1295 No DAC + Ipamorelin** (GH-axis upstream); **BPC-157** / **TB-500** (tissue repair); **Tesamorelin** (visceral fat); **Tirzepatide** (offsets glucose shifts)
Avoid	Insulin or insulin secretagogues (severe compounded hypoglycemia). Exogenous **HGH** simultaneously (redundant downstream signaling). Active malignancy. Diabetic retinopathy
References	peptidedosages · DailyMed · PubMed · ClinicalTrials · Drugs.com

Safety & Patient Education

Contraindications	• Active malignancy • Acute critical illness (post-surgical, ICU): increased mortality reported in trials • Severe / proliferative diabetic retinopathy • Pregnancy and lactation
Warnings	• Intracranial hypertension with somatropin (papilledema, vision changes, headache, nausea, vomiting) • Insulin resistance and new-onset T2DM • Fluid retention, edema, carpal tunnel • Pancreatitis (rare) • IGF-1 elevation above age range carries theoretical malignancy risk • Hypoglycemia (most important, dose-related), intracranial hypertension, lymphoid hyperplasia (tonsillar)
Drug interactions	• Glucocorticoids: blunt GH/IGF-1 response — reassess steroid dose after starting GH • Insulin / antidiabetics: GH raises insulin needs; IGF-1 and IGF-1 LR3 cause hypoglycemia — adjust accordingly • Levothyroxine: GH lowers free T4 — recheck TSH/fT4 quarterly • Oral estrogens: blunt GH-stimulated IGF-1 — consider transdermal if response is blunted • CYP3A4 narrow-TI substrates (cyclosporine, tacrolimus, sirolimus): recheck troughs after starting • Anticonvulsants (phenytoin, carbamazepine, phenobarbital): GH may shift seizure threshold — monitor levels
Labs	• Compound-specific: IGF-1 (baseline + Week 8 + end), Fasting glucose (Week 0/4/8), Lipid panel (Week 0/12) • IGF-1 at baseline and quarterly during cycles • Fasting glucose, HbA1c quarterly • TSH and free T4 quarterly (GH affects thyroid axis) • Cortisol baseline (GH can mask central adrenal insufficiency) • Lipid panel • Hip/scoliosis exam in pediatrics
Stop criteria	• Severe headache with vision changes (rule out intracranial hypertension) • New onset hyperglycemia • Persistent edema • New-onset hip pain in pediatrics
Patient education	• Vials must be refrigerated never frozen (somatropin) • Rotate injection sites • Report headaches with nausea/vision changes immediately • Report new joint pain • Children: report any limp or hip/knee pain immediately • Always have fast-acting carbs available before injection. • Do not inject and skip a meal.

IGF-1 LR3

GROWTH HORMONE AXIS

In plain English	*A modified IGF-1 that hangs around 20-30× longer than the regular version. Used by athletes for sustained growth signaling.*
What it does	Long-acting IGF-1 analog (20-30 hr half-life) - drives muscle protein synthesis and cellular repair directly. Binds IGF-1R with high affinity. Activates PI3K/Akt/mTOR for protein synthesis.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
0.1 mg	0.1mg + 0.2 mL acetic acid	0.5 mg/mL	1 unit = 0.005 mg
1 mg	1mg + 2 mL acetic acid	0.5 mg/mL	1 unit = 0.005 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	Daily post-workout	**50 mcg**	10u	Systemic dose.
Site IM	Post-workout	**25-40 mcg/site**	5-8u	Local muscle.
Off	Equal to on-time	**-**	-	Mandatory.
Cycle length	4-6 weeks on, equal time off. Off periods critical - receptor desensitises permanently if skipped.			
Timing	With food, post-workout — long half-life means hypoglycemia risk all day			

Handling & Stacking

Injection site	IM into recently trained muscle (site enhancement) or SubQ abdomen (systemic).
Storage	Lyophilised: -4°F. Reconstituted in acetic acid: 36-46°F, use within 4 weeks.
Side effects	Hypoglycaemia (keep carbs available). Jaw/finger growth at high doses. Headache, nausea.
What to expect	Muscle fullness within days. Recovery at 2-4 weeks. Hypoglycaemia risk - always have fast carbs available. Off-cycles are mandatory.
Supply math	**0.1mg** → 14 vials | **1mg** → 2 vials *(4-wk @ 50 mcg × 7/wk)*
Pairs well with	**CJC-1295 No DAC + Ipamorelin** (GH-axis support); **Tesamorelin** (visceral fat reduction); **BPC-157** / **TB-500** (joint and soft-tissue repair); **Tirzepatide** (offsets glucose shifts)
Avoid	Active malignancy (promotes growth). Diabetic retinopathy. Take with food (NOT fasted/bedtime - blunts effect)
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox · peptides.id

Safety & Patient Education

Contraindications	• Active malignancy • Acute critical illness (post-surgical, ICU): increased mortality reported in trials • Severe / proliferative diabetic retinopathy • Pregnancy and lactation
Warnings	• Intracranial hypertension with somatropin (papilledema, vision changes, headache, nausea, vomiting) • Insulin resistance and new-onset T2DM • Fluid retention, edema, carpal tunnel • Pancreatitis (rare) • IGF-1 elevation above age range carries theoretical malignancy risk • 20-30 hour half-life means hypoglycemia risk extends throughout the day. • Long-term IGF-1 elevation theoretical malignancy concern.
Drug interactions	• Glucocorticoids: blunt GH/IGF-1 response — reassess steroid dose after starting GH • Insulin / antidiabetics: GH raises insulin needs; IGF-1 and IGF-1 LR3 cause hypoglycemia — adjust accordingly • Levothyroxine: GH lowers free T4 — recheck TSH/fT4 quarterly • Oral estrogens: blunt GH-stimulated IGF-1 — consider transdermal if response is blunted • CYP3A4 narrow-TI substrates (cyclosporine, tacrolimus, sirolimus): recheck troughs after starting • Anticonvulsants (phenytoin, carbamazepine, phenobarbital): GH may shift seizure threshold — monitor levels
Labs	• Compound-specific: IGF-1 (baseline + Week 8 + end), Fasting glucose (Week 0/4/8), Lipid panel (Week 0/12) • IGF-1 at baseline and quarterly during cycles • Fasting glucose, HbA1c quarterly • TSH and free T4 quarterly (GH affects thyroid axis) • Cortisol baseline (GH can mask central adrenal insufficiency) • Lipid panel • Hip/scoliosis exam in pediatrics
Stop criteria	• Severe headache with vision changes (rule out intracranial hypertension) • New onset hyperglycemia • Persistent edema • New-onset hip pain in pediatrics
Patient education	• Vials must be refrigerated never frozen (somatropin) • Rotate injection sites • Report headaches with nausea/vision changes immediately • Report new joint pain • Children: report any limp or hip/knee pain immediately • Eat carbs after injection, not fasted

IPAMORELIN

GROWTH HORMONE AXIS

In plain English	*Cleanest of the GH-releasers. Gives you a natural GH pulse without dragging cortisol or prolactin up with it.*
What it does	Cleanest GHRP - selective GH secretagogue with minimal cortisol or prolactin elevation. Selective GHS-R1a agonist without cortisol, prolactin, or appetite effects. The preferred GHRP for CJC-1295 + Ipamorelin stacks.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
2 mg	2mg + 1 mL	2 mg/mL	1 unit = 0.02 mg
5 mg	5mg + 2.5 mL	2 mg/mL	1 unit = 0.02 mg
10 mg	10mg + 3 mL	3.33 mg/mL	1 unit = 0.03 mg

Protocol

Weeks	Frequency	Dose	Units 2 mg/mL	Units 3.33 mg/mL	Notes
Weeks 1-2	Nightly 5×/week	**200 mcg**	10u	6.0u	Start low and assess how your body responds before increasing.
Weeks 3-12	Nightly 5×/week	**300 mcg**	15u	9.0u	Standard effective dose for most users.
Off	4 weeks	**-**	-	-	Off period - let your pituitary and receptors recover before the next cycle.
Cycle length	8-12 weeks on / 4-8 weeks off. Max: 16 weeks on / 4-8 weeks off. Cleanest GHRP receptor profile but cycling still preserves response.				
Timing	Fasted pre-bed nightly (or fasted AM). No food 60-90 min before; no food 60 min after — cleanest GHRP profile				

Handling & Stacking

Injection site	Abdomen preferred.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, up to 6 weeks.
Side effects	Mild water retention, brief tingling. Minimal cortisol/prolactin vs
What to expect	Improved sleep within 1-2 weeks. Very mild tingling. Body composition changes over 8-12 weeks. No appetite increase - the preferred GHRP for most protocols.
Supply math	**2mg** → 9 vials | **5mg** → 4 vials | **10mg** → 2 vials *(12-wk @ 300 mcg × 5/wk)*
Pairs well with	**CJC-1295** (No DAC) (canonical pair, GH pulse synergy); **Wolverine Blend** (**BPC-157 + TB-500**) (recovery); Mod-GRF (additional GHRH pulse); MK-677 (oral, extends GH window)
Avoid	Active malignancy. Severe insulin resistance
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox · peptides.id

Safety & Patient Education

Contraindications	• Active malignancy of any kind (GH/IGF-1 can promote tumor growth) • Severe diabetic retinopathy • Acute critical illness (per somatropin label — applies to GH-axis stimulators) • Pregnancy and lactation
Warnings	• Insulin resistance and worsening glycemic control with sustained use • Fluid retention, peripheral edema • Carpal tunnel syndrome with extended use • Joint and muscle pain • IGF-1 elevation: theoretical malignancy promotion if dosed beyond physiological IGF-1 ranges • Combined GHRH+GHRP use requires off cycles to prevent receptor desensitization • Cleanest GHRP profile (minimal cortisol/prolactin); class warnings still apply but at lower magnitude
Drug interactions	• Glucocorticoids: blunt GH response — minimize or time away from injection • Insulin / antidiabetics: GH worsens insulin resistance — expect dose adjustments + tighter monitoring • Levothyroxine: GH lowers free T4 — recheck TSH/fT4 quarterly • Oral estrogens: blunt GH-stimulated IGF-1 — transdermal doesn't • CYP3A4 narrow-TI substrates (cyclosporine, tacrolimus, sirolimus): recheck troughs after starting
Labs	• Compound-specific: IGF-1 (baseline + Week 8 + end), Lipid panel (Week 0/12) • IGF-1 at baseline, mid-cycle, and end of cycle. Target ≤ULN for age • Fasting glucose and HbA1c at baseline and end of cycle • Mammogram and prostate screening per age-appropriate guidelines before starting
Stop criteria	• Fasting glucose rises >20 mg/dL above baseline • IGF-1 rises above the upper limit of normal for age • New persistent joint pain or carpal tunnel symptoms • New skin changes or moles enlarging • Vision changes
Patient education	• GH-axis stimulation is not appropriate during active malignancy or for fertility • Most GH pulses occur during deep sleep — dose pre-bed unless instructed otherwise • Food within 90 min of injection blunts GH response (carbs/fats most) — fasted is standard • Report new lumps, persistent headaches, or vision changes immediately

KISSPEPTIN-10

REPRODUCTIVE AND HORMONAL

In plain English	*Tells your pituitary to fire off LH and FSH, which restarts your natural hormone system. Has to be dosed in pulses (like M/W/F) — daily shuts it down.*
What it does	HPG axis activator - stimulates GnRH release to naturally drive testosterone, FSH, and libido through the body's own hormonal casc Activates GPR54 in the hypothalamus, triggering GnRH → LH + FSH surge → testosterone.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 2 mL	2.5 mg/mL	1 unit = 0.03 mg
10 mg	10mg + 3 mL	3.33 mg/mL	1 unit = 0.03 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	M/W/F (pulsatile)	**0.5-1 mcg/kg**	-	~35-70 mcg for 70 kg. Daily dosing causes GPR54 desensitisation per source Section II.
Fertility	M/W/F or pulsatile	**1-2 mcg/kg**	-	HPG restoration. Avoid daily dosing (GPR54 desensitisation).
Off	30 days	**-**	-	Natural HPG reset.
Cycle length	30-day cycles, 30-day off.			
Timing	3×/week SubQ pulsatile — never daily (GPR54 desensitization)			

Handling & Stacking

Injection site	Abdomen or outer thigh.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 30 days.
Side effects	Testosterone surge symptoms. LH/FSH monitoring required.
What to expect	Testosterone and LH/FSH rise within days. Full HPG restoration takes 30+ days. Libido enhancement within 1-2 weeks.
Pairs well with	**HCG** (additive HPG stimulation); **CJC-1295 No DAC + Ipamorelin** (anabolic environment); **BPC-157** / **TB-500** (recovery during cycle); **PT-141** (sexual response synergy)
Avoid	Used too frequently - GPR54 desensitisation. Hormone-sensitive cancers. Pregnancy
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox · peptides.id

Safety & Patient Education

Contraindications	• Hormone-sensitive malignancy (prostate cancer outside intentional ADT, breast cancer, ovarian cancer) • Active thromboembolic disease • Untreated thrombophilia • Pregnancy (except specific fertility indications under specialist supervision) • Polycythemia vera
Warnings	• Estradiol elevation common; aromatase inhibitor may be needed • Thromboembolic risk with elevated estradiol • Ovarian hyperstimulation syndrome (OHSS) in females on HCG: can be life-threatening • Mood changes, gynecomastia in males • HCG raises both T and E2; estradiol monitoring is mandatory • Daily dosing causes GPR54 desensitization; pulsatile dosing only
Drug interactions	• AIs (anastrozole, letrozole): co-administered to manage E2 — over-suppression worsens lipids and bone • SERMs (clomiphene, tamoxifen, enclomiphene): redundant overlap with HCG via central HPG drive • Exogenous testosterone: shuts down LH/FSH — HCG efficacy reduced unless TRT paused or microdosed • Anticoagulants (warfarin, DOACs): elevated E2 raises thromboembolic risk — coordinate if chronic • Insulin / antidiabetics: HCG may shift glucose at high doses — monitor diabetic patients
Labs	• Compound-specific: Total & Free Testosterone (baseline + end), LH + FSH (Day 0/10/30), Semen analysis (baseline + q3 months) • Total + free testosterone, LH, FSH, sensitive E2 at baseline, mid-cycle, end • SHBG and prolactin at baseline • CBC + hematocrit (polycythemia surveillance) • Lipid panel; PSA in men >40 at baseline • Semen analysis baseline + quarterly for fertility protocols • D-dimer if clotting symptoms
Stop criteria	• Hematocrit >54% — donate blood or hold therapy • Estradiol persistently above target despite AI • OHSS symptoms (abdominal distension, weight gain, dyspnea) • Thrombosis symptoms (calf swelling, chest pain, dyspnea, neuro) • New breast tenderness or growth (gynecomastia workup)
Patient education	• Watch for clotting symptoms: leg swelling, calf pain, chest pain, sudden shortness of breath • Estrogen side effects (gynecomastia, mood) manageable with AI under physician guidance • Fertility protocols: 70+ days before semen analysis changes • HCG raises estrogen significantly • 3x/week maximum; avoid daily dosing

KISSPEPTIN-54

REPRODUCTIVE AND HORMONAL

In plain English	*Longer-lasting version of Kisspeptin-10. Usually given as a single shot to trigger ovulation during IVF.*
What it does	Full-length 54-amino acid kisspeptin - the endogenous master HPG axis regulator. Binds KISS1R/GPR54 on GnRH neurons in hypothalamic arcuate nucleus and AVPV to trigger pulsatile GnRH release → LH/FSH → testoster

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 1 mL	5 mg/mL	1 unit = 0.05 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
IVF trigger	Single dose	**6.4 nmol/kg**	-	Clinical IVF protocol. Oocyte maturation trigger vs hCG.
HPG restoration	3× weekly	**100-300 mcg**	2-6u	Community HPG axis stimulation. Lower desensitisation risk than K-10 due to longer natural pulse.
Off	Cycle-dependent	**-**	-	No standard off period - use pulsatile protocol only.
Cycle length	Pulsatile 3× weekly only. Daily dosing causes GPR54 desensitisation.			
Timing	IVF trigger single shot or weekly			

Handling & Stacking

Injection site	SubQ abdomen. IV for IVF protocols (clinical setting only).
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 28 days.
Side effects	Facial flushing, warmth, mild abdominal discomfort (most common). Transient dizziness with IV administration. Generally very well tolerated.
What to expect	LH surge within 30-60 min of injection. Testosterone rise within 2-4 hours. Sustained HPG activation over 3× weekly protocol builds over 4-6 weeks. Used as IVF trigger: oocyte maturation within 36-38 hrs at clinical dose.
Supply math	**5mg** → 1 vial *(4-wk @ 300 mcg × 3/wk)*
Pairs well with	**HCG** (additive HPG stimulation); **CJC-1295 No DAC + Ipamorelin** (anabolic environment); **BPC-157** / **TB-500** (recovery during cycle); **PT-141** (sexual response synergy)
Avoid	Used too frequently - GPR54 desensitisation. Hormone-sensitive cancers. Pregnancy
References	peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Hormone-sensitive malignancy (prostate cancer outside intentional ADT, breast cancer, ovarian cancer) • Active thromboembolic disease • Untreated thrombophilia • Pregnancy (except specific fertility indications under specialist supervision) • Polycythemia vera
Warnings	• Estradiol elevation common; aromatase inhibitor may be needed • Thromboembolic risk with elevated estradiol • Ovarian hyperstimulation syndrome (OHSS) in females on HCG: can be life-threatening • Mood changes, gynecomastia in males • HCG raises both T and E2; estradiol monitoring is mandatory • Longer half-life than K-10; lower desensitization risk but same axis effects
Drug interactions	• AIs (anastrozole, letrozole): co-administered to manage E2 — over-suppression worsens lipids and bone • SERMs (clomiphene, tamoxifen, enclomiphene): redundant overlap with HCG via central HPG drive • Exogenous testosterone: shuts down LH/FSH — HCG efficacy reduced unless TRT paused or microdosed • Anticoagulants (warfarin, DOACs): elevated E2 raises thromboembolic risk — coordinate if chronic • Insulin / antidiabetics: HCG may shift glucose at high doses — monitor diabetic patients
Labs	• Compound-specific: Total & Free Testosterone (baseline + end), LH + FSH (Day 0/10/30) • Total + free testosterone, LH, FSH, sensitive E2 at baseline, mid-cycle, end • SHBG and prolactin at baseline • CBC + hematocrit (polycythemia surveillance) • Lipid panel; PSA in men >40 at baseline • Semen analysis baseline + quarterly for fertility protocols • D-dimer if clotting symptoms
Stop criteria	• Hematocrit >54% — donate blood or hold therapy • Estradiol persistently above target despite AI • OHSS symptoms (abdominal distension, weight gain, dyspnea) • Thrombosis symptoms (calf swelling, chest pain, dyspnea, neuro) • New breast tenderness or growth (gynecomastia workup)
Patient education	• Watch for clotting symptoms: leg swelling, calf pain, chest pain, sudden shortness of breath • Estrogen side effects (gynecomastia, mood) manageable with AI under physician guidance • Fertility protocols: 70+ days before semen analysis changes • HCG raises estrogen significantly

KLOW BLEND

BLENDS AND MIXED PROTOCOLS

In plain English	*GLOW (BPC + TB + GHK-Cu) plus KPV. Adds gut and joint anti-inflammatory coverage. The go-to blend for IBD, Crohn's, or colitis.*
What it does	Four-peptide blend - and angiogenesis. KPV adds MC1R/MC3R anti-inflammation for gut, joints, and skin.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
80 mg	80mg + 5 mL	GHK-Cu 10 mg/mL BPC-157 2 mg/mL TB-500 2 mg/mL KPV 2 mg/mL	1 unit = 0.10 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Week 1	Daily	**1 mg**	10u	Loading. 1 mg GHK-Cu + 0.2 mg each of BPC/TB/KPV per dose.
Week 2	Daily	**1.5 mg**	15u	Ramp. 1.5 mg GHK-Cu + 0.3 mg each of BPC/TB/KPV per dose.
Weeks 3-8	Daily	**2 mg**	20u	Working dose. 2 mg GHK-Cu + 0.4 mg each of BPC/TB/KPV per dose.
Aggressive	Daily (AM + PM split)	**3-4 mg**	30-40u	Bump here ONLY if 2+ weeks at 2 mg with zero injection-site issues AND a specific high-demand goal. Start at 3 mg (= 0.6 mg each of BPC/TB/KPV). Push to 4 mg only if 3 mg is producing visible results and you want to test further - no published benefit above 3 mg/day; cost roughly doubles.
Off	4 weeks	-	-	Mandatory copper-accumulation washout. Do not skip - especially after an Aggressive phase.
Cycle length	8 weeks on / 4 weeks off - matches GHK-Cu copper-accumulation limits.			
Timing	Daily SubQ, anytime — same as GLOW base			

Handling & Stacking

Injection site	SubQ near target tissue. Rotate.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, protected from light, 6 weeks (rate-limited by GHK-Cu component).
Side effects	Usually minimal.
What to expect	Broader anti-inflammatory coverage than GLOW - preferred for IBD, Crohn's, joint inflammation, and post-surgical recovery. KPV adds gut/inflammatory coverage. 8-week cycle, 4-week washout.
Supply math	**80mg** → 2 vials *(8-wk @ 2 mg × 7/wk)* *80 mg vial covers ~half of an 8-week cycle at 2 mg/day; reconstituted shelf life ~28 days. Aggressive 3-4 mg/day depletes the vial faster but stays inside the discard window.*
Pairs well with	**CJC-1295 No DAC + Ipamorelin** (systemic GH + local repair); **Thymosin Alpha-1** (immune amplification)
Avoid	NSAIDs (antagonise prostaglandin pathway, blunting the blend's repair signalling). High-dose vitamin C same session (copper oxidation)
References	peptidefox · researchdosing · peptidedosages · PubMed · thepeptidereport · peptides.id

Safety & Patient Education

Contraindications	• Active malignancy (TB-500 angiogenesis; BPC-157 tissue growth) • Wilson's disease, copper dysregulation, copper allergy (GHK-Cu component) • Alpha-MSH allergy (KPV is an alpha-MSH C-terminal fragment) • Active bleeding or recent hemorrhagic stroke (TB-500 vascular effects) • Pregnancy, lactation
Warnings	• Four-compound blend with no human RCT data on the combination. • GHK-Cu: copper accumulation risk with extended cycling — 4-6 week off period is mandatory. • BPC-157: FDA Category 2 in 2023; compounding restricted. • TB-500: theoretical malignancy concern via angiogenesis; no published human trials. • KPV: limited human data; oral route is community-standard for gut indications but not RCT-supported.
Drug interactions	• NSAIDs: rodent data suggests blunted BPC-157 effect; clinical relevance uncertain in humans • High-dose vitamin C same session: theoretical copper redox interaction (GHK-Cu) • Anticoagulants: caution with TB-500 vascular effects • Other alpha-MSH agonists (Melanotan I/II): theoretical receptor overlap with KPV
Labs	• Compound-specific: hsCRP (baseline + end) • Serum copper and ceruloplasmin if cycles extend beyond 6 weeks
Stop criteria	• New mass or unexpected tissue growth at any injection site • Skin discoloration or neurological symptoms (copper toxicity) • New skin pigmentation changes (KPV alpha-MSH activity) • Persistent injection-site reactions beyond 7 days
Patient education	• Four-peptide blend: any single-component contraindication applies to the whole blend. • Preferred over GLOW for IBD, Crohn's, ulcerative colitis, and post-surgical gut healing — KPV adds gut/inflammatory coverage. • Cycle discipline: 4-6 weeks on / 4-6 weeks off. Continuous use without washout risks copper accumulation. • Inject SubQ near target tissue. Rotate sites to reduce GHK-Cu sting.

KPV

TISSUE REPAIR AND RECOVERY

In plain English	*Short anti-inflammatory peptide. Works on gut conditions like Crohn's and colitis. Unusual for a peptide — you can take it orally.*
What it does	Alpha-MSH C-terminal tripeptide - potent gut and systemic anti-inflammatory via MC1R/MC3R. Effective orally for GI conditions. Tripeptide (Lys-Pro-Val) from alpha-MSH. Binds MC1R/MC3R to suppress NF-κB, IL-6, TNF-α, and promote gut barrier integrity.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 1 mL	5 mg/mL	1 unit = 0.05 mg
10 mg	10mg + 2 mL	5 mg/mL	1 unit = 0.05 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Oral	Daily	**0.5-2 mg**	10-40u	Gut-targeted.
SubQ	Daily	**0.5-1 mg**	10-20u	Systemic.
Off	2-4 weeks	**-**	-	Essential break between cycles. Skipping leads to tolerance and diminishing returns.
Cycle length	4-6 weeks on / 2-4 weeks off. Max: 8 weeks on / 4 weeks off.			
Timing	Daily, anytime — oral or SubQ			

Handling & Stacking

Injection site	SubQ abdomen.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 4 weeks.
Side effects	Generally very well tolerated. Rare: mild injection-site reactions.
What to expect	Gut symptom improvement within 1-2 weeks orally. Systemic anti-inflammatory via SubQ within days. Very well tolerated.
Supply math	**5mg** → 11 vials | **10mg** → 6 vials *(4-wk @ 2 mg × 7/wk)*
Pairs well with	**BPC-157** (gut healing stack); **TB-500** (tissue repair); LL-37 (additive immune modulation); Larazotide (gut barrier complement)
Avoid	Allergy to alpha-MSH analogues. Pregnancy/nursing. **KLOW Blend** (already contains **KPV** - would double-dose)
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox · peptides.id

Safety & Patient Education

Contraindications	• Alpha-MSH allergy, pregnancy, lactation
Warnings	• Limited human data — oral form for gut conditions is community-standard but not RCT-supported
Drug interactions	• Other alpha-MSH agonists (MT-1, MT-2, PT-141): theoretical receptor overlap — redundant rather than additive • Immunosuppressants (cyclosporine, tacrolimus, mycophenolate): KPV's anti-inflammatory effect may blunt titration markers — coordinate monitoring • Biologics for IBD (infliximab, adalimumab, ustekinumab, vedolizumab): no human data — KPV used as community adjunct but shouldn't replace prescribed therapy
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Stop criteria	• New skin pigmentation changes, persistent headache
Patient education	• Aseptic technique: alcohol swab the vial septum and the injection site, use a single-use sterile needle for every injection, never share vials or needles between people • Sharps disposal in an FDA-cleared sharps container, never household trash • Cold chain: if a refrigerated peptide reaches room temperature for more than the labeled stability window, discard it • Mark the date of first puncture on every multi-dose vial • Inspect every vial before injection: clear solution (unless labeled otherwise), no particulates, no cloudiness, no color change • Rotate injection sites to avoid lipoatrophy, induration, and infection • Carry a record of what you're injecting if you need emergency care

L-CARNITINE

FAT LOSS AND METABOLISM

In plain English	*Shuttles fat into the cell's furnace so it can actually get burned. Helpful when you're in a calorie deficit or training a lot.*
What it does	Amino acid derivative essential for mitochondrial fatty acid transport. Injectable provides 5-10× higher plasma levels than oral. Transfers long-chain fatty acids across the inner mitochondrial membrane via CPT1/CPT2 carnitine shuttle.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
600 mg	600mg + 0.60 mL	1000 mg/mL	1 unit = 10.00 mg
1200 mg	1200mg + 1.2 mL	1000 mg/mL	1 unit = 10.00 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	M/W/F	**500-1,000 mg**	50-100u	Community dose.
High	M/W/F	**1,200-1,500 mg**	120-150u	Fat oxidation range.
Pre-WO	Before exercise	**500 mg**	50u	Timing-specific.
Cycle length	8-12 weeks on / 4 weeks off. Max: ongoing maintenance with quarterly 2-week breaks.			
Timing	Pre-workout (preferred, 30-60 min before) or daily anytime			

Handling & Stacking

Injection site	IM preferred or SubQ. Upper outer thigh for volume.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 30 days.
Side effects	Fishy odour (TMAO conversion). Generally well tolerated.
What to expect	Exercise performance and fat utilisation improvement within 2-4 weeks. Reduced post-workout fatigue. Works best in caloric deficit or with GLP-1 therapy.
Supply math	**600mg** → 59 vials | **1200mg** → 29 vials *(8-wk @ 1500 mg × 3/wk)*
Pairs well with	GLP-1 agonists (fat oxidation during caloric deficit); **LC526** / **LC216** (lipotropic stack); **NAD+** (mitochondrial stack)
Avoid	Hypothyroid medication - separate by 4 hrs. Warfarin (may potentiate)
References	researchdosing · peptidedosages · DailyMed · PubMed · ClinicalTrials · Drugs.com

Safety & Patient Education

Contraindications	• Component allergies (B12, riboflavin, lecithin, choline, methionine, lidocaine) • Hypothyroid medication — separate by ≥4 hours • Severe hepatic impairment (choline metabolism) • Pregnancy and lactation — specific compounding required • Bleeding disorders or anticoagulants (relative) — bruising risk • Severe renal impairment (carnitine accumulation) • Seizure disorder (case reports of seizure exacerbation) • Hypothyroidism without dose separation
Warnings	• Compounded products — composition varies; request the pharmacy's COA • B12 (cyanocobalamin) contains cobalt — allergy possible • Methionine elevates homocysteine in some individuals • L-Carnitine: TMAO conversion concern in CV disease (controversial — baseline lipids) • Injection-site reactions common with phosphatidylcholine • TMAO conversion may be relevant in patients with cardiovascular disease. • Fishy odor common.
Drug interactions	• Warfarin: L-carnitine and choline blends can potentiate — monitor INR if combining • Levothyroxine: separate by ≥4 hours — choline/B-vitamins affect absorption • Anticoagulants / antiplatelets (DOACs, aspirin, clopidogrel): bruising risk with phosphatidylcholine • Insulin / antidiabetics: B-complex may shift glucose — recheck log if dosing changes • Methotrexate, trimethoprim, phenytoin: deplete folate / B12 — coordinate with prescriber • Lidocaine blends (LC526): avoid other caines same session — additive cardiac effects • Stimulants / phentermine: additive CV load with high-dose lipotropics — monitor BP/HR • Warfarin (potentiation), thyroid medication (separate by 4 hours)
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required • Baseline B12, folate, MMA, homocysteine (methylation markers) • TSH at baseline (B12 deficiency masks hypothyroidism) • Liver enzymes (choline metabolism) • Hematocrit (B12 raises RBC — polycythemia risk in some)
Stop criteria	• New seizure activity
Patient education	• Fishy odor from choline/L-carnitine is normal — not allergic • Inject slowly to reduce stinging • Don't use with warfarin without physician guidance (potentiation risk) • Request exact ingredient list and concentrations from your compounder

LC120

BLENDS AND MIXED PROTOCOLS

In plain English	*Classic Lipo-C: methionine + choline + L-carnitine + B5 (dexpanthenol). No B12. Empower-style formula.*
What it does	Lipo-C - classic lipotropic blend: Methionine + Choline + L-Carnitine + Dexpanthenol (B5). No B12. Empower-style formula. Methionine + Choline: hepatic VLDL export. L-Carnitine: mitochondrial fat oxidation.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 ml	Solution no reconstitution needed	Methionine 15 mg/mL Choline Cl 50 mg/mL L-Carnitine 50 mg/mL Dexpanthenol 5 mg/mL	1 unit = 0.15 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	1-3×/week	**0.5-1 mL**	50-100u	IM upper outer glute preferred; SubQ acceptable. 1 mL/week is most common maintenance.
Off	2-4 weeks	**-**	-	Reassess at 16-20 weeks. No mandatory off if tolerated.
Cycle length	Ongoing 8-12 week blocks. Reassess at 16-20 weeks.			
Timing	3×/week IM, anytime — typically AM			

Handling & Stacking

Injection site	IM upper outer glute preferred; SubQ acceptable.
Storage	Solution: refrigerate 36-46°F. Use within 30 days. Protect from light.
Side effects	Garlic/sulfur taste post-injection (methionine), mild flushing, injection-site sting.
What to expect	Energy and fat-loss adjunct effect builds over 4-8 weeks alongside diet/exercise.
Pairs well with	GLP-1s (fat-loss adjunct); **L-Carnitine** standalone (additive); **SS-31** / **NAD+** (mitochondrial)
Avoid	Sulfa allergy (theoretical methionine).
References	peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Component allergies (B12, riboflavin, lecithin, choline, methionine, lidocaine) • Hypothyroid medication — separate by ≥4 hours • Severe hepatic impairment (choline metabolism) • Pregnancy and lactation — specific compounding required • Bleeding disorders or anticoagulants (relative) — bruising risk
Warnings	• Compounded products — composition varies; request the pharmacy's COA • B12 (cyanocobalamin) contains cobalt — allergy possible • Methionine elevates homocysteine in some individuals • L-Carnitine: TMAO conversion concern in CV disease (controversial — baseline lipids) • Injection-site reactions common with phosphatidylcholine
Drug interactions	• Warfarin: L-carnitine and choline blends can potentiate — monitor INR if combining • Levothyroxine: separate by ≥4 hours — choline/B-vitamins affect absorption • Anticoagulants / antiplatelets (DOACs, aspirin, clopidogrel): bruising risk with phosphatidylcholine • Insulin / antidiabetics: B-complex may shift glucose — recheck log if dosing changes • Methotrexate, trimethoprim, phenytoin: deplete folate / B12 — coordinate with prescriber • Lidocaine blends (LC526): avoid other caines same session — additive cardiac effects • Stimulants / phentermine: additive CV load with high-dose lipotropics — monitor BP/HR
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required • Baseline B12, folate, MMA, homocysteine (methylation markers) • TSH at baseline (B12 deficiency masks hypothyroidism) • Liver enzymes (choline metabolism) • Hematocrit (B12 raises RBC — polycythemia risk in some)
Patient education	• Fishy odor from choline/L-carnitine is normal — not allergic • Inject slowly to reduce stinging • Don't use with warfarin without physician guidance (potentiation risk) • Request exact ingredient list and concentrations from your compounder

LC216

BLENDS AND MIXED PROTOCOLS

In plain English	*Lipo-B: extended lipotropic with carnitine + arginine + MIC + B6 + B5 + B12. The most-prescribed clinic lipotropic.*
What it does	Lipo-B - extended lipotropic: MIC + L-Carnitine + L-Arginine + B6 + B5 + B12. Methionine + Choline + Inositol: hepatic VLDL export. L-Carnitine: mitochondrial fat oxidation. L-Arginine: NO precursor.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 ml	Solution no reconstitution needed	L-Carnitine 20 mg/mL L-Arginine 20 mg/mL Methionine 25 mg/mL Inositol 50 mg/mL Choline 50 mg/mL B6 25 mg/mL B5 25 mg/mL B12 1 mg/mL	1 unit = 0.20 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-4	Weekly	**1 mL**	100u	Start at 1 mL weekly to assess tolerance. AgelessRx standard onboarding.
Weeks 5+	1-2×/week	**1 mL**	100u	Increase to 2 mL/week or 1 mL twice weekly once tolerated. Energy and fat-loss adjunct dose.
Off	2-4 weeks	-	-	Optional break between 12-week blocks.
Cycle length	Ongoing 8-12 week blocks; reassess at 16-20 weeks.			
Timing	3×/week IM, anytime — typically AM			

Handling & Stacking

Injection site	IM upper outer glute preferred; SubQ acceptable.
Storage	Solution: refrigerate 36-46°F. Use within 30 days. Protect from light.
Side effects	Garlic taste (methionine), flushing (niacin family), injection-site sting.
What to expect	Energy lift within 1-2 weeks; fat-loss adjunct effect over 8-12 weeks.
Pairs well with	GLP-1s (fat-loss adjunct); **L-Carnitine** standalone (additive)
Avoid	Sulfa allergy (theoretical methionine).
References	peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Component allergies (B12, riboflavin, lecithin, choline, methionine, lidocaine) • Hypothyroid medication — separate by ≥4 hours • Severe hepatic impairment (choline metabolism) • Pregnancy and lactation — specific compounding required • Bleeding disorders or anticoagulants (relative) — bruising risk
Warnings	• Compounded products — composition varies; request the pharmacy's COA • B12 (cyanocobalamin) contains cobalt — allergy possible • Methionine elevates homocysteine in some individuals • L-Carnitine: TMAO conversion concern in CV disease (controversial — baseline lipids) • Injection-site reactions common with phosphatidylcholine
Drug interactions	• Warfarin: L-carnitine and choline blends can potentiate — monitor INR if combining • Levothyroxine: separate by ≥4 hours — choline/B-vitamins affect absorption • Anticoagulants / antiplatelets (DOACs, aspirin, clopidogrel): bruising risk with phosphatidylcholine • Insulin / antidiabetics: B-complex may shift glucose — recheck log if dosing changes • Methotrexate, trimethoprim, phenytoin: deplete folate / B12 — coordinate with prescriber • Lidocaine blends (LC526): avoid other caines same session — additive cardiac effects • Stimulants / phentermine: additive CV load with high-dose lipotropics — monitor BP/HR
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required • Baseline B12, folate, MMA, homocysteine (methylation markers) • TSH at baseline (B12 deficiency masks hypothyroidism) • Liver enzymes (choline metabolism) • Hematocrit (B12 raises RBC — polycythemia risk in some)
Patient education	• Fishy odor from choline/L-carnitine is normal — not allergic • Inject slowly to reduce stinging • Don't use with warfarin without physician guidance (potentiation risk) • Request exact ingredient list and concentrations from your compounder

LC526

BLENDS AND MIXED PROTOCOLS

In plain English	*Heaviest lipotropic — MIC + 300 mg/mL carnitine + B12 + B6 + NADH. For fat-loss-focused users on calorie restriction.*
What it does	Lipo-C Fat Blaster - high-concentration MIC + L-Carnitine + B12 + B6 + NADH. Highest mg/mL lipotropic on the market. 300 mg/mL L-Carnitine exceeds the threshold for measurable mitochondrial fat-transport enhancement.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 ml	Solution no reconstitution needed	L-Carnitine 300 mg/mL Methionine 25 mg/mL Inositol 50 mg/mL Choline 50 mg/mL B12 1 mg/mL B6 50 mg/mL NADH 50 mg/mL	1 unit = 3.00 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	1-2×/week	**0.5 mL**	50u	Half-volume vs LC120 because concentration is ~4× higher (526 mg/mL total). IM upper outer glute preferred.
Off	2-4 weeks	-	-	Reassess at 12-16 weeks for ongoing use.
Cycle length	8-12 weeks on / 2-4 off. Reassess at 16 weeks.			
Timing	3×/week IM, anytime — lidocaine reduces injection sting			

Handling & Stacking

Injection site	IM upper outer glute. SubQ tolerable but stings at high carnitine concentration.
Storage	Solution: refrigerate 36-46°F. Use within 30 days. Protect from light (NADH photodegrades).
Side effects	Stronger garlic taste (high carnitine), flushing (B6/NADH), injection-site sting (300 mg/mL carnitine).
What to expect	Stronger fat-loss adjunct effect than LC120/LC216 over 8-12 weeks. Energy improvement within days.
Pairs well with	GLP-1s (fat-loss adjunct); **SS-31** + **NAD+** (mitochondrial energy stack)
Avoid	Sulfa allergy (theoretical methionine). Sensitive injection sites.
References	peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Component allergies (B12, riboflavin, lecithin, choline, methionine, lidocaine) • Hypothyroid medication — separate by ≥4 hours • Severe hepatic impairment (choline metabolism) • Pregnancy and lactation — specific compounding required • Bleeding disorders or anticoagulants (relative) — bruising risk
Warnings	• Compounded products — composition varies; request the pharmacy's COA • B12 (cyanocobalamin) contains cobalt — allergy possible • Methionine elevates homocysteine in some individuals • L-Carnitine: TMAO conversion concern in CV disease (controversial — baseline lipids) • Injection-site reactions common with phosphatidylcholine • Lidocaine component contraindicated in lidocaine allergy and significant cardiac conduction disease
Drug interactions	• Warfarin: L-carnitine and choline blends can potentiate — monitor INR if combining • Levothyroxine: separate by ≥4 hours — choline/B-vitamins affect absorption • Anticoagulants / antiplatelets (DOACs, aspirin, clopidogrel): bruising risk with phosphatidylcholine • Insulin / antidiabetics: B-complex may shift glucose — recheck log if dosing changes • Methotrexate, trimethoprim, phenytoin: deplete folate / B12 — coordinate with prescriber • Lidocaine blends (LC526): avoid other caines same session — additive cardiac effects • Stimulants / phentermine: additive CV load with high-dose lipotropics — monitor BP/HR
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required • Baseline B12, folate, MMA, homocysteine (methylation markers) • TSH at baseline (B12 deficiency masks hypothyroidism) • Liver enzymes (choline metabolism) • Hematocrit (B12 raises RBC — polycythemia risk in some)
Patient education	• Fishy odor from choline/L-carnitine is normal — not allergic • Inject slowly to reduce stinging • Don't use with warfarin without physician guidance (potentiation risk) • Request exact ingredient list and concentrations from your compounder

LIPO-C

FAT LOSS AND METABOLISM

In plain English	*A vitamin-and-amino-acid injection blend (MIC + L-Carnitine + B12 or B5). Two compounding styles: Empower swaps B12 for Dexpanthenol (B5); Olympia and AgelessRx keep B12. Check your supplier's label.*
What it does	Lipotropic phosphatidylcholine injection - hepatic fat mobilisation, liver health, and metabolic support. Phosphatidylcholine is the primary hepatic fat transport molecule.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	Solution no reconstitution needed	Phosphatidylcholine-based formulation	U-100 syringe

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	M/W/F	**1 mL**	100u	Phosphatidylcholine lipotropic injection.
High	M/W/F	**2 mL**	200u	Upper range - only if you tolerated lower doses well.
Cycle length	Ongoing or 8-12 weeks during weight loss protocols.			
Timing	Daily or 3×/week, anytime — typically AM for energy bump			

Handling & Stacking

Injection site	Abdomen or upper outer thigh.
Storage	Solution: refrigerate. Use within 28 days.
Side effects	Injection-site stinging (phosphatidylcholine). Fishy odour.
What to expect	Liver function and fat metabolism improvement over 4-8 weeks. Works best alongside caloric restriction and GLP-1 therapy.
Pairs well with	GLP-1 agonists (amplified fat loss); **CJC-1295 No DAC + Ipamorelin** (full body recomp); **Vitamin B12** (energy + mood); **Tesamorelin** (visceral fat)
Avoid	No significant contraindications
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Component allergies (B12, riboflavin, lecithin, choline, methionine, lidocaine) • Hypothyroid medication — separate by ≥4 hours • Severe hepatic impairment (choline metabolism) • Pregnancy and lactation — specific compounding required • Bleeding disorders or anticoagulants (relative) — bruising risk
Warnings	• Compounded products — composition varies; request the pharmacy's COA • B12 (cyanocobalamin) contains cobalt — allergy possible • Methionine elevates homocysteine in some individuals • L-Carnitine: TMAO conversion concern in CV disease (controversial — baseline lipids) • Injection-site reactions common with phosphatidylcholine
Drug interactions	• Warfarin: L-carnitine and choline blends can potentiate — monitor INR if combining • Levothyroxine: separate by ≥4 hours — choline/B-vitamins affect absorption • Anticoagulants / antiplatelets (DOACs, aspirin, clopidogrel): bruising risk with phosphatidylcholine • Insulin / antidiabetics: B-complex may shift glucose — recheck log if dosing changes • Methotrexate, trimethoprim, phenytoin: deplete folate / B12 — coordinate with prescriber • Lidocaine blends (LC526): avoid other caines same session — additive cardiac effects • Stimulants / phentermine: additive CV load with high-dose lipotropics — monitor BP/HR
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required • Baseline B12, folate, MMA, homocysteine (methylation markers) • TSH at baseline (B12 deficiency masks hypothyroidism) • Liver enzymes (choline metabolism) • Hematocrit (B12 raises RBC — polycythemia risk in some)
Patient education	• Fishy odor from choline/L-carnitine is normal — not allergic • Inject slowly to reduce stinging • Don't use with warfarin without physician guidance (potentiation risk) • Request exact ingredient list and concentrations from your compounder

MELANOTAN I (MT-1)

REPRODUCTIVE AND HORMONAL

In plain English	*Triggers a tan. FDA-approved as Scenesse for a rare condition where sunlight is painful. More selective and safer than MT-2.*
What it does	FDA-approved melanocortin agonist (Scenesse) - selective MC1R activation increases melanin and provides photoprotection. Selective MC1R agonist driving melanogenesis. More selective than MT-2 - minimal MC4R effects.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 2 mL	5 mg/mL	1 unit = 0.05 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Loading	Daily × 2 weeks	**0.5-1 mg**	10-20u	Build melanin.
Maintenance	2×/week	**1 mg**	20u	Sustain pigmentation.
Off	With cycle	**-**	-	As desired.
Cycle length	Loading: 2 weeks daily. Maintenance: 2× weekly.			
Timing	Daily during loading, then maintenance 2-3×/week			

Handling & Stacking

Injection site	SubQ abdomen or outer thigh.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 4 weeks. Protect from light.
Side effects	Nausea (lower than MT-2), flushing, hyperpigmentation.
What to expect	Melanin increases within days. Noticeable tan over 2-4 weeks. Less nausea than MT-2. Baseline skin check recommended before starting.
Supply math	**10mg** → 1 vial *(4-wk @ 1 mg × 2/wk)*
Pairs well with	Vitamin D (offsets some pigmentation cofactor demand)
Avoid	Personal or family history of melanoma. Active dysplastic naevi (mole monitoring required)
References	researchdosing · peptidedosages · DailyMed · PubMed · ClinicalTrials · Drugs.com

Safety & Patient Education

Contraindications	• Personal or family history of melanoma • Atypical mole syndrome / dysplastic nevus syndrome • Active skin cancer or pre-malignant skin lesions • Uncontrolled hypertension • Cardiovascular disease (significant) • Pregnancy and lactation
Warnings	• Hyperpigmentation of all melanocytes including existing moles — mole monitoring required • New mole formation • Transient BP elevation (PT-141: mean ~6 mmHg systolic per FDA Vyleesi label) • Nausea, especially MT-2 during loading • Priapism risk in men (MT-2 specifically) • Spontaneous erections (MT-2) • Reduced response and tachyphylaxis with frequent use • More selective for MC1R; less nausea, less libido effect, less BP impact than MT-2; same melanoma surveillance still applies
Drug interactions	• PDE5 inhibitors (sildenafil, tadalafil, vardenafil): additive BP effects with PT-141 — separate timing per label • Antihypertensives: additive or paradoxical BP effects • Vasoactive drugs in general
Labs	• Compound-specific: Dermatology review - mole mapping baseline + ABCDE q12 weeks during active use • Dermatology baseline mole mapping with photographic documentation • Repeat dermatology review every 12 weeks during active use, every 6-12 months after • Blood pressure log at baseline and before each dose
Stop criteria	• New atypical mole or rapid change in existing mole (ABCDE criteria) • Sustained BP elevation above 140/90 in previously normotensive patient • Priapism (sustained erection more than 4 hours): seek emergency care • Severe nausea not resolving with dose reduction
Patient education	• Use sunscreen during MT-1/MT-2 — extra melanin doesn't replace UV protection • Photograph moles at baseline; recheck monthly • ABCDE warning signs: Asymmetry, Border irregularity, Color variation, Diameter >6mm, Evolution • MT-2: erections are pharmacologic — >4 hours is a medical emergency • PT-141: do not exceed 1 dose per 24 hours

MELANOTAN II (MT-2)

REPRODUCTIVE AND HORMONAL

In plain English	*Gives you a deep tan. Side effects include nausea, a libido boost, and unprompted erections. Keep an eye on your moles — they darken too.*
What it does	Non-selective melanocortin agonist - increases melanin (tanning) and libido via MC4R. Parent compound of PT-141. Activates MC1R (melanogenesis), MC3R (energy), and MC4R (sexual arousal).

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 3 mL	3.33 mg/mL	1 unit = 0.03 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Load Week 1	Nightly	**167 mcg**	5.0u	Start low.
Load Week 2	Nightly	**333 mcg**	10u	Step up.
Maintenance	1-2×/week	**500 mcg**	15.0u	Sustain pigmentation.
Cycle length	2-week loading phase then ongoing maintenance.			
Timing	Daily SubQ during loading; anytime; libido use 1-2 hr before activity			

Handling & Stacking

Injection site	SubQ abdomen. Rotate.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 4 weeks. Protect from light.
Side effects	Flushing/nausea (dose-dependent, resolves). Darkening of all pigmented lesions. Libido increase.
What to expect	Skin darkening within days. Flushing and nausea expected at loading doses - resolve with titration. Spontaneous erections in men. Monitor all moles - pigmentation increases globally.
Supply math	**10mg** → 1 vial *(2-wk @ 500 mcg × 2/wk)*
Pairs well with	**PT-141** (libido/erectile complement); GH secretagogues (skin regeneration); **B12** (energy + nausea support)
Avoid	Personal or family history of melanoma. Atypical mole syndrome. Recent prolonged unprotected sun exposure
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox · peptides.id

Safety & Patient Education

Contraindications	• Personal or family history of melanoma • Atypical mole syndrome / dysplastic nevus syndrome • Active skin cancer or pre-malignant skin lesions • Uncontrolled hypertension • Cardiovascular disease (significant) • Pregnancy and lactation
Warnings	• Hyperpigmentation of all melanocytes including existing moles — mole monitoring required • New mole formation • Transient BP elevation (PT-141: mean ~6 mmHg systolic per FDA Vyleesi label) • Nausea, especially MT-2 during loading • Priapism risk in men (MT-2 specifically) • Spontaneous erections (MT-2) • Reduced response and tachyphylaxis with frequent use • Priapism risk in men. Spontaneous erections expected. Greater pigmentation magnitude than MT-1.
Drug interactions	• PDE5 inhibitors (sildenafil, tadalafil, vardenafil): additive BP effects with PT-141 — separate timing per label • Antihypertensives: additive or paradoxical BP effects • Vasoactive drugs in general
Labs	• Compound-specific: Dermatology review - mole mapping baseline + ABCDE q12 weeks during active use • Dermatology baseline mole mapping with photographic documentation • Repeat dermatology review every 12 weeks during active use, every 6-12 months after • Blood pressure log at baseline and before each dose
Stop criteria	• New atypical mole or rapid change in existing mole (ABCDE criteria) • Sustained BP elevation above 140/90 in previously normotensive patient • Priapism (sustained erection more than 4 hours): seek emergency care • Severe nausea not resolving with dose reduction • Erection lasting more than 4 hours: emergency care
Patient education	• Use sunscreen during MT-1/MT-2 — extra melanin doesn't replace UV protection • Photograph moles at baseline; recheck monthly • ABCDE warning signs: Asymmetry, Border irregularity, Color variation, Diameter >6mm, Evolution • MT-2: erections are pharmacologic — >4 hours is a medical emergency • PT-141: do not exceed 1 dose per 24 hours

MIC BLEND

FAT LOSS AND METABOLISM

In plain English	*A classic 'lipotropic' shot — methionine, inositol, choline, and B12. Used 2-3× a week as a fat-loss helper.*
What it does	Classic MIC + B12 lipotropic - Methionine + Inositol + Choline + B12 for hepatic fat metabolism and energy. Methionine: fat transport and detox. Inositol: lipid metabolism and insulin signalling. Choline: hepatic VLDL export.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	Solution no reconstitution needed	Methionine 25 mg/mL Inositol 50 mg/mL Choline 50 mg/mL B12 1 mg/mL	1 unit = 0.25 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	M/W/F	**1 mL**	100u	Methionine 25 + Inositol 50 + Choline 50 + B12 1 mg/mL.
Daily	Daily	**0.5 mL**	50u	Lower daily alternative.
Cycle length	Ongoing with weight loss protocols.			
Timing	Daily or 3×/week, anytime — typically AM			

Handling & Stacking

Injection site	SubQ or IM upper outer thigh.
Storage	Solution: refrigerate. Use within 30 days.
Side effects	Mild injection-site reactions. Fishy odour (choline).
What to expect	Energy from B12 rapid. Fat metabolism improvement over 4-8 weeks. Standard lipotropic adjunct with GLP-1 therapy.
Pairs well with	GLP-1 agonists (lipotropic stack); **LC120** / **LC216** / **LC526** (**Lipo-C** variants); **HHB** (B-vitamins)
Avoid	B-vitamin allergy. Severe hepatic disease (choline metabolism)
References	peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Component allergies (B12, riboflavin, lecithin, choline, methionine, lidocaine) • Hypothyroid medication — separate by ≥4 hours • Severe hepatic impairment (choline metabolism) • Pregnancy and lactation — specific compounding required • Bleeding disorders or anticoagulants (relative) — bruising risk
Warnings	• Compounded products — composition varies; request the pharmacy's COA • B12 (cyanocobalamin) contains cobalt — allergy possible • Methionine elevates homocysteine in some individuals • L-Carnitine: TMAO conversion concern in CV disease (controversial — baseline lipids) • Injection-site reactions common with phosphatidylcholine
Drug interactions	• Warfarin: L-carnitine and choline blends can potentiate — monitor INR if combining • Levothyroxine: separate by ≥4 hours — choline/B-vitamins affect absorption • Anticoagulants / antiplatelets (DOACs, aspirin, clopidogrel): bruising risk with phosphatidylcholine • Insulin / antidiabetics: B-complex may shift glucose — recheck log if dosing changes • Methotrexate, trimethoprim, phenytoin: deplete folate / B12 — coordinate with prescriber • Lidocaine blends (LC526): avoid other caines same session — additive cardiac effects • Stimulants / phentermine: additive CV load with high-dose lipotropics — monitor BP/HR
Labs	• Compound-specific: Liver enzymes AST/ALT (baseline) - baseline hepatic function before starting • Baseline B12, folate, MMA, homocysteine (methylation markers) • TSH at baseline (B12 deficiency masks hypothyroidism) • Liver enzymes (choline metabolism) • Hematocrit (B12 raises RBC — polycythemia risk in some)
Patient education	• Fishy odor from choline/L-carnitine is normal — not allergic • Inject slowly to reduce stinging • Don't use with warfarin without physician guidance (potentiation risk) • Request exact ingredient list and concentrations from your compounder

MOTS-C

FAT LOSS AND METABOLISM

In plain English	*Tricks your cells into acting like you just exercised. People use it for energy, fat-burning, and better insulin response.*
What it does	Mitochondria-derived peptide that activates AMPK for metabolic health, energy, and insulin sensitivity. Encoded in 12S rRNA of mtDNA.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 0.50 mL	20 mg/mL	1 unit = 0.20 mg
40 mg	40mg + 2 mL	20 mg/mL	1 unit = 0.20 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	M/W/F AM	**5 mg**	25u	Working dose. Exercise-mimetic - pair with training days if possible.
Off	4 weeks	**-**	-	Essential break. Skipping leads to tolerance and diminishing returns.
Cycle length	4-6 weeks on (5 mg 3×/week) / 4 weeks off. The Peptide Report Performance range; aligned with the upper bound of ResearchDosing community protocols.			
Timing	Fasted AM (M/W/F) or fasted 30-60 min pre-workout. No food 60-90 min before; no food 60 min after — exercise-mimetic; avoid evening doses			

Handling & Stacking

Injection site	Abdomen or outer thigh.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 7 days (strict stability limit).
Side effects	Early fatigue, appetite suppression, occasional nausea - resolves after week 1.
What to expect	Early AMPK-driven fatigue and appetite suppression in weeks 1-2 - expected and resolves. Improved metabolic function builds over the 8-week cycle. Monitor fasting glucose when stacked with 5-Amino and NAD+.
Supply math	**10mg** → 5 vials | **40mg** → 2 vials *(4-wk @ 5 mg × 3/wk)*
Pairs well with	**5-Amino-1MQ** (**NAD** preservation pairs with mitochondrial restoration); **NAD+** (mitochondrial stack); **SS-31** (additive mitochondrial protection); **Tirzepatide** (metabolic preservation during weight loss)
Avoid	No major contraindications documented
References	peptidedosingprotocols · researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Pregnancy, lactation. • Active malignancy.
Warnings	• No human RCT data. • AMPK activation overlaps with metformin and berberine.
Drug interactions	• Metformin, berberine (additive glucose lowering)
Labs	• Compound-specific: Fasting glucose (Week 0/4/8), HbA1c (baseline + end)
Stop criteria	• Persistent hypoglycemia
Patient education	• Aseptic technique: alcohol swab the vial septum and the injection site, use a single-use sterile needle for every injection, never share vials or needles between people • Sharps disposal in an FDA-cleared sharps container, never household trash • Cold chain: if a refrigerated peptide reaches room temperature for more than the labeled stability window, discard it • Mark the date of first puncture on every multi-dose vial • Inspect every vial before injection: clear solution (unless labeled otherwise), no particulates, no cloudiness, no color change • Rotate injection sites to avoid lipoatrophy, induration, and infection • Carry a record of what you're injecting if you need emergency care

NAD+

SUPPLEMENTS AND COFACTORS

In plain English	*A coenzyme that powers your cells and drops as you age. The shot delivers higher levels than oral precursors can. Inject slowly so you don't flush.*
What it does	Master cellular energy coenzyme - declines 50% between age 20 and 50. Injectable achieves levels impossible with oral precursors. Coenzyme in 500+ enzyme reactions. Substrate for sirtuins (epigenetic repair) and PARP (DNA repair). Powers ATP production.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
500 mg	500mg + 2 mL	250 mg/mL	1 unit = 2.50 mg
1000 mg	1000mg + 4 mL	250 mg/mL	1 unit = 2.50 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Week 1	M/W/F	**25 mg**	10u	Tolerance ramp. Start gentle to assess flushing and chest-pressure tolerance. Inject slowly over 30-60 seconds.
Week 2	M/W/F	**50 mg**	20u	Step up if Week 1 was tolerated. Continue slow injection.
Week 3	M/W/F	**75 mg**	30u	Step up if Week 2 was tolerated.
Weeks 4+	M/W/F	**100 mg**	40u	Working dose (300 mg/week - aligned with Empower/Olympia/Tailor Made compounded SubQ ceiling). Hold at lower step if headache, chest pressure, or fatigue developed at the prior step.
Advanced	M-F (5×/week)	**75-100 mg**	30-40u	Optional escalation after the working dose is well tolerated for at least 4 weeks. 375-500 mg/week - above the standard compounded ceiling but within research-community range. Step back to 3×/week if any new flushing, headache, fatigue, or chest pressure appears.
Off	4 weeks	**-**	-	Recovery cycle for sirtuin/PARP balance. Quarterly 2-week breaks acceptable for ongoing chronic use.
Cycle length	8-12 weeks on / 4 weeks off. Max: ongoing with quarterly 2-week breaks. Cycling supports sirtuin/PARP balance.			
Timing	3×/week SubQ, slow injection (30-60 sec) — AM preferred (alerting)			

Handling & Stacking

Injection site	Outer thigh preferred for volume tolerance.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, maximum 14 days - firm limit. One vial at a time.
Side effects	Flushing, stimulation, insomnia if dosed late. Nausea at higher doses.
What to expect	Stimulating warmth and energy during/after injection - normal. Cognitive clarity and metabolic improvement over 4-8 weeks. 14-day stability limit after reconstitution.
Supply math	**500mg** → 2 vials | **1000mg** → 1 vial *(4-wk @ 100 mg × 3/wk)*
Pairs well with	**MOTS-c** (mitochondrial stack); **SS-31** (additive mitochondrial protection); **5-Amino-1MQ** (**NAD**-preserving NNMT inhibition); **Glutathione** (antioxidant); Resveratrol (sirtuin activation)
Avoid	Active migraine (**NAD+** infusions can trigger). Severe cardiovascular disease (BP fluctuations)
References	olympiapharmacy · agelessrx · empower · researchdosing · peptidedosages · PubMed

Safety & Patient Education

Contraindications	• Active migraine (NAD+ infusions can trigger), severe cardiovascular disease (BP fluctuations)
Warnings	• Stimulating effect can disrupt sleep if dosed late. • Inject slowly to avoid acute flushing/chest pressure.
Drug interactions	• Antihypertensives and vasoactive medications: NAD+ administration can cause transient BP changes and chest pressure — monitor BP, especially with rapid push or large doses • Stimulants (caffeine high-dose, ADHD medications): additive activation; reduce dose or shift timing if combined • Sedatives/sleep medications: NAD+ is alerting and counteracts sedation; avoid evening dosing if combining • Methylating agents (TMG, methylfolate, methylcobalamin): theoretical complementary pathway support but no human data; some users titrate methyl donors when NAD+ doses are high • Chemotherapy: NAD+ supplementation in active cancer treatment is contested — coordinate with oncology before starting
Labs	• Compound-specific: Liver enzymes AST/ALT (baseline) - monitor closely if >200 mg/day • Liver enzymes if dosed >200 mg/day
Stop criteria	• Persistent headache, chest pressure during injection
Patient education	• Inject slowly over 30-60 seconds. • Don't dose evening if sleep disruption occurs.

PT-141 (BREMELANOTIDE)

REPRODUCTIVE AND HORMONAL

In plain English	*Vyleesi. FDA-approved for low sex drive in women. Works on arousal in the brain, not blood flow. Take it 30-90 minutes before sex.*
What it does	FDA-approved melanocortin agonist for HSDD - activates MC4R in the brain to increase sexual desire and arousal centrally. Activates MC4R in hypothalamic arousal circuits. Unlike PDE5 inhibitors (blood flow), PT-141 works on the brain.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 1 mL	10 mg/mL	1 unit = 0.10 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
FDA dose women	As-needed	**1.75 mg**	17.5u	FDA-approved Vyleesi dose.
Community men	As-needed	**1-2 mg**	10-20u	Off-label research.
Max	-	**1×/day, 8×/month**	-	FDA Vyleesi label: max 1 dose per 24 hours AND max 8 doses per month. Exceeding 8/month carries significantly higher focal-hyperpigmentation risk (see Side effects).
Cycle length	PRN (situational) per FDA Vyleesi label. Max 1 dose per 24 hours; max 8 doses per month. Not for daily or scheduled use.			
Timing	Single shot 30-90 min before sexual activity (situational)			

Handling & Stacking

Injection site	Abdomen SubQ.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 30 days.
Side effects	Nausea, flushing, BP elevation (transient). Less nausea PM vs AM. Focal hyperpigmentation (face/gums/breasts): ~1% with intermittent PRN use, but 38% when dosed daily for 8 consecutive days (FDA Vyleesi label) - argues strongly against scheduled daily dosing.
What to expect	Sexual desire and arousal within 30-60 min. Duration 6-24 hours. Nausea and flushing common - less if dosed in the evening. Use caution if combining with PDE5 inhibitors (sildenafil, tadalafil) - additive transient BP and HR effects; not an FDA-labeled contraindication.
Pairs well with	**Melanotan II (MT-2)** (libido + tan); **Kisspeptin-10** (HPG axis support)
Avoid	Uncontrolled hypertension (transient BP rise on injection). PDE5 inhibitors at peak (additive transient BP elevation). Cardiovascular disease
References	accessdata.fda · defymedical · empower · researchdosing · peptidedosages · DailyMed

Safety & Patient Education

Contraindications	• Personal or family history of melanoma • Atypical mole syndrome / dysplastic nevus syndrome • Active skin cancer or pre-malignant skin lesions • Uncontrolled hypertension • Cardiovascular disease (significant) • Pregnancy and lactation
Warnings	• Hyperpigmentation of all melanocytes including existing moles — mole monitoring required • New mole formation • Transient BP elevation (PT-141: mean ~6 mmHg systolic per FDA Vyleesi label) • Nausea, especially MT-2 during loading • Priapism risk in men (MT-2 specifically) • Spontaneous erections (MT-2) • Reduced response and tachyphylaxis with frequent use • Focal hyperpigmentation reported • Nausea most pronounced AM, less PM
Drug interactions	• PDE5 inhibitors (sildenafil, tadalafil, vardenafil): additive BP effects with PT-141 — separate timing per label • Antihypertensives: additive or paradoxical BP effects • Vasoactive drugs in general • Naltrexone: decreases naltrexone exposure significantly • Narrow-TI oral medications: delayed absorption
Labs	• Compound-specific: Blood pressure before each dose - stop if SBP drop >20 mmHg from baseline • Dermatology baseline mole mapping with photographic documentation • Repeat dermatology review every 12 weeks during active use, every 6-12 months after • Blood pressure log at baseline and before each dose • BP before each dose
Stop criteria	• New atypical mole or rapid change in existing mole (ABCDE criteria) • Sustained BP elevation above 140/90 in previously normotensive patient • Priapism (sustained erection more than 4 hours): seek emergency care • Severe nausea not resolving with dose reduction • Sustained BP elevation, persistent hyperpigmentation
Patient education	• Use sunscreen during MT-1/MT-2 — extra melanin doesn't replace UV protection • Photograph moles at baseline; recheck monthly • ABCDE warning signs: Asymmetry, Border irregularity, Color variation, Diameter >6mm, Evolution • MT-2: erections are pharmacologic — >4 hours is a medical emergency • PT-141: do not exceed 1 dose per 24 hours • Max 8 doses/month per FDA Vyleesi label.

RETATRUTIDE

FAT LOSS AND METABOLISM

In plain English	*Hits three receptors at once (GLP-1, GIP, glucagon). So far the strongest weight-loss drug ever tested in humans — about 28% loss in late trials. Not yet FDA-approved.*
What it does	Triple GLP-1/GIP/Glucagon agonist - most powerful weight loss compound in trials. Simultaneously activates GLP-1R (appetite), GIPR (insulin + fat), and GcgR (thermogenesis + hepatic fat).

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 0.50 mL	20 mg/mL	1 unit = 0.20 mg
15 mg	15mg + 0.75 mL	20 mg/mL	1 unit = 0.20 mg
20 mg	20mg + 1 mL	20 mg/mL	1 unit = 0.20 mg
30 mg	30mg + 1.5 mL	20 mg/mL	1 unit = 0.20 mg
60 mg	60mg + 3 mL	20 mg/mL	1 unit = 0.20 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-4	1× weekly	**0.5 mg/week**	2.5u	Entry. Tolerance only - NOT for weight loss. Sub-5u draw; use a 0.3 mL precision syringe.
Weeks 5-8	1× weekly	**1 mg/week**	5u	Adapt. First real appetite effect. Step up only if Weeks 1-4 were asymptomatic.
Weeks 9-12	1× weekly	**2 mg/week**	10u	Trial dose. Target floor - many first-timers stop here.
Weeks 13-16	1× weekly	**4 mg/week**	20u	Trial dose. Common sweet spot. Goal = minimum effective dose.
Weeks 17-24	1× weekly	**8 mg/week**	40u	Trial dose. Step up only if 4 mg is failing.
Weeks 25+	1× weekly	**12 mg/week**	60u	TRIUMPH Phase 3 ceiling. Refractory cases under specialist supervision only.
Cycle length	First-time-user protocol targets 4-6 mg/week (not 12 mg). Goal = minimum effective dose, not maximum tolerable. Hold at the lowest dose that controls appetite and produces results. Phase 3 TRIUMPH-4 (Dec 2025 topline) ran 68 weeks continuous at fixed-titration doses 4-12 mg/week; long-term safety beyond 68 weeks not established.			
Timing	Weekly (any day) — same day each week; time of day flexible			

Handling & Stacking

Injection site	Abdomen, outer thigh, or upper arm.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 28 days.
Side effects	Nausea, vomiting, decreased appetite, diarrhea. GI more pronounced than dual agonists at high doses.
What to expect	Aggressive appetite suppression from week 1. GI side effects more intense than semaglutide during titration. 12 mg arm achieved 28.7% weight loss at 68 weeks (9 mg: 26.4%) in TRIUMPH-4 Phase 3 topline (Dec 2025). Full publication pending.
Supply math	**15mg** → 19 vials | **20mg** → 14 vials | **30mg** → 10 vials | **60mg** → 5 vials *(25-wk @ 12 mg × 1/wk)*
Pairs well with	**MOTS-c** (mitochondrial support during rapid weight loss); **L-Carnitine** (fatty-acid mitochondrial transport)
Avoid	**Semaglutide** or **Tirzepatide** simultaneously, or any other GLP-1 receptor agonist (additive GI side effects, GLP-1 desensitisation). Insulin without monitoring
References	pmc · peptidefox · researchdosing · peptidedosages · PubMed · thepeptidereport

RETATRUTIDE

FAT LOSS AND METABOLISM — SAFETY & PATIENT EDUCATION

Contraindications	• Personal or family history of medullary thyroid carcinoma (MTC) or MEN 2 • Personal or family history of any thyroid cancer • Personal history of pancreatitis (any cause) • Pre-existing gastroparesis or significant gastric motility disorder • Severe GERD, scleroderma, or other conditions with delayed gastric emptying • Prior serious hypersensitivity to the compound or excipients • Pregnancy and lactation; discontinue ≥2 months (sema) or ≥1 month (tirz) before planned conception • History of cholelithiasis or gallbladder disease • Active or pre-existing diabetic retinopathy • Eating disorder history (rapid weight loss can trigger relapse) • Chronic kidney disease stage 4-5 • History of any cancer (slowed gastric motility complicates oral chemotherapy absorption)
Warnings	• Acute kidney injury from GI-related volume depletion — risk highest during dose escalation • Acute pancreatitis: discontinue if suspected (severe abdominal pain radiating to back) • Acute gallbladder disease (cholelithiasis, cholecystitis) • Diabetic retinopathy progression in patients with pre-existing retinopathy • Hypoglycemia with insulin or sulfonylureas — reduce those at GLP-1 initiation • Pulmonary aspiration during anesthesia. Hold ≥1 wk (weekly) or day-of (daily) per ASA • Severe GI: ileus, obstruction, severe constipation, fecal impaction • Anaphylaxis and angioedema reported postmarketing • Tachycardia: small mean HR increases (1-4 bpm) reported • Triple-agonist: GI tolerance significantly worse than dual or single agonists at high doses. • Heart rate elevation can be more pronounced. • Glucagon component raises glucose excursions.
Drug interactions	• Delayed gastric emptying affects oral absorption (levothyroxine, oral contraceptives, warfarin) — separate timing • Insulin / secretagogues: dose reduction at GLP-1 initiation • Other GLP-1 agonists: do not combine
Pregnancy / Lactation	• Semaglutide: stop ≥2 mo before planned pregnancy (long half-life). Tirzepatide: stop ≥1 mo per Mounjaro label.
Labs	• Compound-specific: BMP/CMP (creatinine, eGFR - baseline + dose escalation), Fasting glucose (Week 0/4/8), HbA1c (baseline + end), Lipid panel (Week 0/12), Lipase (if abdominal pain) • Baseline: BMP/CMP, CBC, HbA1c, lipid panel, TSH, pregnancy test (childbearing potential) • During therapy: BMP at every dose escalation, and any time GI side effects cause >24h reduced food/fluid intake • HbA1c at 3 + 6 mo; lipid panel at 6 mo; repeat eGFR if signs of dehydration • Drop hsCRP and routine calcitonin (not standard of care) • Resting heart rate, fasting glucose more frequent, lipid panel
Stop criteria	• Creatinine rise >0.3 mg/dL or eGFR drop >20%: hold, rehydrate, reassess • Severe abdominal pain: hold and evaluate for pancreatitis • Vision changes: ophthalmology evaluation • Persistent vomiting >48h: hold dose • New neck mass, hoarseness, dysphagia, dyspnea: thyroid pathology workup
Patient education	• AKI on GLP-1s is from dehydration, not the drug. Target 2.5-3 L water/day; add electrolytes if low PO. Hold next dose for any day of vomiting/diarrhea/reduced food; resume same level after 24h recovery. Same-day contact for: decreased urination, dark urine, flank pain, ankle swelling, confusion. • Required during any GLP-1 cycle: protein ≥1.2 g/kg IBW/day (1.6 if training), resistance training 2-3×/wk, creatine 5 g/day. Without this, 25-40% of weight lost is lean mass and regain is rapid. • Eat slowly, smaller portions, stop at first fullness • Same-day medical contact: severe abdominal pain (especially radiating to back), persistent vomiting, decreased urination, dark urine, vision changes, severe constipation, neck mass, hoarseness • Inform any anesthesiologist before procedures (aspiration risk) • Use contraception during therapy and washout • Pen sharing prohibited even with new needle • Phase 3 only; long-term safety beyond 68 weeks not established. Not FDA-approved. • Compounded versions are not equivalent to clinical trial product.

SELANK

COGNITIVE AND NEUROPROTECTIVE

In plain English	*Russian anti-anxiety peptide. Calms you down without making you drowsy.*
What it does	Anxiolytic nootropic - Russian pharmaceutical for anxiety and cognitive enhancement. Approved in Russia for anxiety disorders. Tuftsin analog modulating GABAergic and serotonergic systems. Inhibits enkephalin-degrading enzymes. Stimulates BDNF and NGF.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 2 mL	2.5 mg/mL	1 unit = 0.03 mg
10 mg	10mg + 3 mL	3.33 mg/mL	1 unit = 0.03 mg

Protocol

Weeks	Frequency	Dose	Units 2.5 mg/mL	Units 3.33 mg/mL	Notes
Intranasal	Daily AM	**250-500 mcg**	2.5-5 sprays	2.5-5 sprays	Dissolve and spray.
SubQ	Daily AM	**250-500 mcg**	10-20u	7.5-15.0u	Systemic alternative.
Off	2 weeks	**-**	-	-	Prevents tolerance.
Cycle length	4-6 weeks on, 2 weeks off.				
Timing	Daily SubQ — anytime; calmer than Semax. See Selank (Intranasal) for the nasal-route page.				

Handling & Stacking

Injection site	Intranasal (primary). SubQ abdomen (alternative).
Storage	Lyophilised or solution: refrigerate. Use within 30 days.
Side effects	Mild sedation or relaxation. Rarely: headache. Generally very well tolerated.
What to expect	Anxiolytic effect within 20-30 min of intranasal. Clear-headed calmness - not sedating. Memory and focus improvement over 2-4 weeks. No addiction or withdrawal.
Supply math	**5mg** → 3 vials | **10mg** → 2 vials *(4-wk @ 500 mcg × 7/wk)*
Pairs well with	**Semax** (synergistic nootropic / mood support); **BPC-157** (neuroregeneration); **GHK-Cu** (neuroprotective complement); L-Theanine (calming amplification)
Avoid	MAOIs (theoretical serotonergic interaction). Active psychosis / bipolar mania
References	innerbody · researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Active psychosis or mania • MAOI therapy (theoretical serotonergic interaction) • Pregnancy and lactation
Warnings	• Approved in Russia / Ukraine / some CIS states — not FDA-approved • Quality varies between research-chemical and pharmaceutical-grade suppliers • Headache common in initial days (Semax) — often responds to choline
Drug interactions	• MAOIs: avoid • SSRIs/SNRIs: theoretical serotonergic risk — monitor for serotonin syndrome • Stimulants: additive activation — reduce dose if combined • Sedatives/benzodiazepines: Selank may augment GABA effects
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required • No specific markers — baseline CBC + CMP
Patient education	• Neuroactive compounds — effects are real and dosing matters • Don't combine with prescription antidepressants without physician input • Russian clinical practice: 10-14 day courses, not continuous

SELANK (INTRANASAL)

COGNITIVE AND NEUROPROTECTIVE

In plain English	*Selank as a nasal spray.*
What it does	Anxiolytic nootropic - Russian pharmaceutical for anxiety and cognitive enhancement. Approved in Russia for anxiety disorders. Tuftsin analog modulating GABAergic and serotonergic systems. Inhibits enkephalin-degrading enzymes. Stimulates BDNF and NGF.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 5 mL sterile saline	1 mg/mL	1 spray = 100 mcg (100 µL)
10 mg	10mg + 10 mL sterile saline	1 mg/mL	1 spray = 100 mcg (100 µL) (10 mL nasal spray bottle)

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	Daily AM	**200-400 mcg**	2-4 sprays	1-2 sprays per nostril, alternating sides. 100 µL @ 1 mg/mL = 100 mcg per spray.
Higher	Daily AM	**600 mcg**	6.0 sprays	3 sprays per nostril, alternating sides. Or split AM + early afternoon to reduce nasal irritation.
Off	1-2 weeks	**-**	-	Tolerance prevention. 5-day-on / 2-day-off variant also community-standard.
Cycle length	4-6 weeks on, 2 weeks off.			
Timing	AM intranasal — calibrate sprays with the supplier's nasal pump (standard 100 µL per actuation) before clinical use.			

Handling & Stacking

Injection site	Intranasal: 1-2 sprays per nostril, alternating sides each dose. Use a calibrated nasal pump (100 µL/spray, 100 mcg per spray at this recon) and tilt head slightly back during instillation. For SubQ injection, see the Selank page.
Storage	Lyophilised: -4°F. Reconstituted in saline: 36-46°F, use within 21 days. Discard if cloudy or particulate.
Side effects	Mild sedation or relaxation. Rarely: headache. Generally very well tolerated.
What to expect	Anxiolytic effect within 20-30 min of intranasal. Clear-headed calmness - not sedating. Memory and focus improvement over 2-4 weeks. No addiction or withdrawal.
Supply math	**5mg** → 4 vials | **10mg** → 2 vials *(4-wk @ 600 mcg × 7/wk)*
Pairs well with	**Semax** (synergistic nootropic / mood support); **BPC-157** (neuroregeneration); **GHK-Cu** (neuroprotective complement); L-Theanine (calming amplification)
Avoid	MAOIs (theoretical serotonergic interaction). Active psychosis / bipolar mania
References	PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Patient education	• Aseptic technique: alcohol swab the vial septum and the injection site, use a single-use sterile needle for every injection, never share vials or needles between people • Sharps disposal in an FDA-cleared sharps container, never household trash • Cold chain: if a refrigerated peptide reaches room temperature for more than the labeled stability window, discard it • Mark the date of first puncture on every multi-dose vial • Inspect every vial before injection: clear solution (unless labeled otherwise), no particulates, no cloudiness, no color change • Rotate injection sites to avoid lipoatrophy, induration, and infection • Carry a record of what you're injecting if you need emergency care

SEMAGLUTIDE

FAT LOSS AND METABOLISM

In plain English	*Ozempic and Wegovy. Weekly shot that slows your stomach and quiets hunger. The drug that started the GLP-1 wave.*
What it does	FDA-approved GLP-1 receptor agonist (Ozempic®/Wegovy®) - appetite suppression via hypothalamic satiety centres, gastric emptying d GLP-1 receptor agonist with 94% amino acid sequence homology to native GLP-1.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 2 mL	5 mg/mL	1 unit = 0.05 mg
15 mg	15mg + 3 mL	5 mg/mL	1 unit = 0.05 mg
20 mg	20mg + 4 mL	5 mg/mL	1 unit = 0.05 mg
50 mg	50mg + 10 mL	5 mg/mL	1 unit = 0.05 mg (reconstitute into 10 mL BAC water vial)

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-4	1× weekly	**0.25 mg/week**	5u	FDA starting dose. At U-100 precision floor; draw carefully. Hold for 4 weeks - most GI side effects occur early.
Weeks 5-8	1× weekly	**0.5 mg/week**	10u	Step up only if Weeks 1-4 tolerated. Many users find sufficient appetite control here long-term (Ozempic standard).
Weeks 9-12	1× weekly	**1 mg/week**	20u	Mid-titration. Sweet spot for many users. Ozempic max is 2 mg/week; further steps below are Wegovy-only.
Weeks 13-16	1× weekly	**1.7 mg/week**	34u	FDA Wegovy intermediate step. Step up only if 1 mg appetite control is fading.
Weeks 17+	1× weekly	**2.4 mg/week**	48u	FDA Wegovy maintenance maximum. Stop at any earlier rung that controls appetite without intolerable side effects.
Cycle length	Ongoing per FDA Wegovy / Ozempic labels. No scheduled off-cycle. Discontinue at least 2 months before planned pregnancy (long half-life).			
Timing	Weekly (any day) — same day each week; time of day flexible			

Handling & Stacking

Injection site	Abdomen, outer thigh, or upper arm.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F. See vial label.
Side effects	Nausea, vomiting, diarrhea, constipation, decreased appetite.
What to expect	Appetite reduction within week 1. Nausea peaks weeks 1-4 then diminishes. Weight loss most apparent at 12-16 weeks. Titrate slowly to minimise GI side effects.
Supply math	**10mg** → 5 vials | **15mg** → 3 vials | **20mg** → 3 vials | **50mg** → 1 vial *(17-wk @ 2.4 mg × 1/wk)*
Pairs well with	**MOTS-c** (mitochondrial energy support during caloric deficit); **Lipo-C** (lipotropic adjunct); **Vitamin B12** (counters GLP-1 **B12** depletion)
Avoid	**Tirzepatide** or **Retatrutide** simultaneously, or any other GLP-1 receptor agonist (GLP-1 receptor desensitisation + additive GI nausea, hypoglycaemia)
References	rxlist · researchdosing · peptidedosages · DailyMed · PubMed · ClinicalTrials

SEMAGLUTIDE

FAT LOSS AND METABOLISM — SAFETY & PATIENT EDUCATION

Contraindications	• Personal or family history of medullary thyroid carcinoma (MTC) or MEN 2 • Personal or family history of any thyroid cancer • Personal history of pancreatitis (any cause) • Pre-existing gastroparesis or significant gastric motility disorder • Severe GERD, scleroderma, or other conditions with delayed gastric emptying • Prior serious hypersensitivity to the compound or excipients • Pregnancy and lactation; discontinue ≥2 months (sema) or ≥1 month (tirz) before planned conception • History of cholelithiasis or gallbladder disease • Active or pre-existing diabetic retinopathy • Eating disorder history (rapid weight loss can trigger relapse) • Chronic kidney disease stage 4-5 • History of any cancer (slowed gastric motility complicates oral chemotherapy absorption)
Warnings	• Acute kidney injury from GI-related volume depletion — risk highest during dose escalation • Acute pancreatitis: discontinue if suspected (severe abdominal pain radiating to back) • Acute gallbladder disease (cholelithiasis, cholecystitis) • Diabetic retinopathy progression in patients with pre-existing retinopathy • Hypoglycemia with insulin or sulfonylureas — reduce those at GLP-1 initiation • Pulmonary aspiration during anesthesia. Hold ≥1 wk (weekly) or day-of (daily) per ASA • Severe GI: ileus, obstruction, severe constipation, fecal impaction • Anaphylaxis and angioedema reported postmarketing • Tachycardia: small mean HR increases (1-4 bpm) reported
Drug interactions	• Delayed gastric emptying affects oral absorption (levothyroxine, oral contraceptives, warfarin) — separate timing • Insulin / secretagogues: dose reduction at GLP-1 initiation • Other GLP-1 agonists: do not combine
Pregnancy / Lactation	• Semaglutide: stop ≥2 mo before planned pregnancy (long half-life). Tirzepatide: stop ≥1 mo per Mounjaro label.
Labs	• Compound-specific: BMP/CMP (creatinine, eGFR - baseline + dose escalation), Fasting glucose (Week 0/4/8), HbA1c (baseline + end), Lipid panel (Week 0/12), Lipase (if abdominal pain) • Baseline: BMP/CMP, CBC, HbA1c, lipid panel, TSH, pregnancy test (childbearing potential) • During therapy: BMP at every dose escalation, and any time GI side effects cause >24h reduced food/fluid intake • HbA1c at 3 + 6 mo; lipid panel at 6 mo; repeat eGFR if signs of dehydration • Drop hsCRP and routine calcitonin (not standard of care)
Stop criteria	• Creatinine rise >0.3 mg/dL or eGFR drop >20%: hold, rehydrate, reassess • Severe abdominal pain: hold and evaluate for pancreatitis • Vision changes: ophthalmology evaluation • Persistent vomiting >48h: hold dose • New neck mass, hoarseness, dysphagia, dyspnea: thyroid pathology workup
Patient education	• AKI on GLP-1s is from dehydration, not the drug. Target 2.5-3 L water/day; add electrolytes if low PO. Hold next dose for any day of vomiting/diarrhea/reduced food; resume same level after 24h recovery. Same-day contact for: decreased urination, dark urine, flank pain, ankle swelling, confusion. • Required during any GLP-1 cycle: protein ≥1.2 g/kg IBW/day (1.6 if training), resistance training 2-3×/wk, creatine 5 g/day. Without this, 25-40% of weight lost is lean mass and regain is rapid. • Eat slowly, smaller portions, stop at first fullness • Same-day medical contact: severe abdominal pain (especially radiating to back), persistent vomiting, decreased urination, dark urine, vision changes, severe constipation, neck mass, hoarseness • Inform any anesthesiologist before procedures (aspiration risk) • Use contraception during therapy and washout • Pen sharing prohibited even with new needle • Hold dose for at least 1 week before any procedure requiring general anesthesia or deep sedation (anesthesia guidance varies; check current ASA position) • Use contraception; discontinue at least 2 months before planned pregnancy • Pen sharing prohibited even with needle change

SEMAX

COGNITIVE AND NEUROPROTECTIVE

In plain English	*Russian cognitive peptide. People use it for focus, memory, and handling stress better. It's stimulating, so use it in the morning.*
What it does	Approved pharmaceutical in Russia and Ukraine - and stroke recovery. analog stimulating BDNF/NGF synthesis, upregulating serotonin/dopamine receptors, and activating PDGF for neuroprotection.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 1 mL	5 mg/mL	1 unit = 0.05 mg
10 mg	10mg + 2 mL	5 mg/mL	1 unit = 0.05 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	Daily AM	**500 mcg**	10u	Russian standard for cognitive use. SubQ abdomen or thigh.
Higher	Daily AM	**1,000 mcg**	20u	Research-range. Some users split into AM + early afternoon doses.
Off	1-2 weeks	**-**	-	Tolerance prevention. 5-day-on / 2-day-off variant also community-standard.
Cycle length	2-4 week blocks, 1-2 weeks off.			
Timing	AM SubQ — stimulating; avoid afternoon/evening doses. See Semax (Intranasal) for the nasal-route page.			

Handling & Stacking

Injection site	SubQ abdomen, upper outer thigh, or back of upper arm. For intranasal route, see the Semax (Intranasal) page.
Storage	Lyophilised: -4°F. Reconstituted/solution: refrigerate, use within 30 days.
Side effects	Headache (first 1-3 days, resolves). Irritability if stacked with stimulants.
What to expect	Mental clarity and focus often within hours of first dose. Sustained cognitive enhancement builds over 2-4 weeks. Brief headache in days 1-3 is normal.
Supply math	**5mg** → 6 vials | **10mg** → 3 vials *(4-wk @ 1000 mcg × 7/wk)*
Pairs well with	**Selank** (anxiolytic + cognitive synergy); **BPC-157** (neurorepair); Alpha-GPC or Citicoline (choline source prevents headache)
Avoid	MAOIs. Active psychosis / mania
References	innerbody · researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Active psychosis or mania • MAOI therapy (theoretical serotonergic interaction) • Pregnancy and lactation
Warnings	• Approved in Russia / Ukraine / some CIS states — not FDA-approved • Quality varies between research-chemical and pharmaceutical-grade suppliers • Headache common in initial days (Semax) — often responds to choline
Drug interactions	• MAOIs: avoid • SSRIs/SNRIs: theoretical serotonergic risk — monitor for serotonin syndrome • Stimulants: additive activation — reduce dose if combined • Sedatives/benzodiazepines: Selank may augment GABA effects
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required • No specific markers — baseline CBC + CMP
Patient education	• Neuroactive compounds — effects are real and dosing matters • Don't combine with prescription antidepressants without physician input • Russian clinical practice: 10-14 day courses, not continuous • Headache common in days 1-3; choline source (alpha-GPC, citicoline) helps

SEMAX (INTRANASAL)

COGNITIVE AND NEUROPROTECTIVE

In plain English	*Semax as a nasal spray.*
What it does	Approved pharmaceutical in Russia and Ukraine - and stroke recovery. analog stimulating BDNF/NGF synthesis, upregulating serotonin/dopamine receptors, and activating PDGF for neuroprotection.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 5 mL sterile saline	1 mg/mL	1 spray = 100 mcg (100 µL)
10 mg	10mg + 10 mL sterile saline	1 mg/mL	1 spray = 100 mcg (100 µL) (10 mL nasal spray bottle)

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	Daily AM	**200-400 mcg**	2-4 sprays	1-2 sprays per nostril, alternating sides. 100 µL @ 1 mg/mL = 100 mcg per spray.
Higher	Daily AM	**600 mcg**	6.0 sprays	3 sprays per nostril, alternating sides. Or split AM + early afternoon to reduce nasal irritation.
Off	1-2 weeks	**-**	-	Tolerance prevention. 5-day-on / 2-day-off variant also community-standard.
Cycle length	2-4 week blocks, 1-2 weeks off.			
Timing	AM intranasal — calibrate sprays with the supplier's nasal pump (standard 100 µL per actuation) before clinical use.			

Handling & Stacking

Injection site	Intranasal: 1-2 sprays per nostril, alternating sides each dose. Use a calibrated nasal pump (100 µL/spray, 100 mcg per spray at this recon) and tilt head slightly back during instillation. For SubQ injection, see the Semax page.
Storage	Lyophilised: -4°F. Reconstituted in saline: 36-46°F, use within 21 days. Discard if cloudy or particulate.
Side effects	Headache (first 1-3 days, resolves). Irritability if stacked with stimulants.
What to expect	Mental clarity and focus often within hours of first dose. Sustained cognitive enhancement builds over 2-4 weeks. Brief headache in days 1-3 is normal.
Supply math	**5mg** → 4 vials | **10mg** → 2 vials *(4-wk @ 600 mcg × 7/wk)*
Pairs well with	**Selank** (anxiolytic + cognitive synergy); **BPC-157** (neurorepair); Alpha-GPC or Citicoline (choline source prevents headache)
Avoid	MAOIs. Active psychosis / mania
References	PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Patient education	• Aseptic technique: alcohol swab the vial septum and the injection site, use a single-use sterile needle for every injection, never share vials or needles between people • Sharps disposal in an FDA-cleared sharps container, never household trash • Cold chain: if a refrigerated peptide reaches room temperature for more than the labeled stability window, discard it • Mark the date of first puncture on every multi-dose vial • Inspect every vial before injection: clear solution (unless labeled otherwise), no particulates, no cloudiness, no color change • Rotate injection sites to avoid lipoatrophy, induration, and infection • Carry a record of what you're injecting if you need emergency care

SHB

BLENDS AND MIXED PROTOCOLS

In plain English	*Super Human Blend — 9-component amino acid + NAC pre-workout injection for energy, performance, and recovery.*
What it does	Super Human Blend - amino acid and NO-precursor injectable for energy, endurance, and recovery. 9-component formula. L-Arginine + L-Ornithine + L-Citrulline: NO precursors for blood flow. L-Carnitine: mitochondrial fat oxidation.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 ml	Solution no reconstitution needed	L-Arg 11 mg/mL L-Orn 11 mg/mL L-Cit 12 mg/mL L-Lys 7 mg/mL L-Gln 4 mg/mL L-Pro 6 mg/mL L-Tau 6 mg/mL L-Car 22 mg/mL NAC 7.5 mg/mL	1 unit = 0.11 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Pre-workout	3-5×/week	**1 mL**	100u	Inject 30-60 min before exercise. Full amino + NAC dose for endurance and recovery.
Daily AM	Daily fasted AM	**0.5 mL**	50u	Lower-dose maintenance variant - half-vial daily AM fasted.
Off	1-2 weeks	**-**	-	Optional washout between 8-12 week performance blocks.
Cycle length	8 weeks on / 4 weeks off. Max: 12 weeks on / 4-6 weeks off. Component-specific tolerance varies.			
Timing	3×/week SubQ/IM, anytime			

Handling & Stacking

Injection site	Upper outer thigh or abdomen. Rotate.
Storage	Solution: refrigerate 36-46°F. Use within 30 days.
Side effects	Mild stimulation (arginine, carnitine). Possible flushing.
What to expect	Energy and pump within 30-60 min pre-workout. Cumulative metabolic and recovery benefits over 4-8 weeks.
Pairs well with	GLP-1 agents (metabolic complement); **NAD+** (mitochondrial energy); **L-Carnitine** standalone (additive)
Avoid	Late PM dosing (stimulating amino acids).
References	peptidedosages · PubMed · thepeptidereport · peptidefox

Safety & Patient Education

Contraindications	• Component allergies, herpes outbreak (high-dose arginine), severe renal/hepatic impairment
Warnings	• Composition varies by supplier
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Patient education	• Verify exact composition and concentrations

SS-31 (ELAMIPRETIDE)

FAT LOSS AND METABOLISM

In plain English	*Protects the tiny power plants inside your cells from age-related damage. Recently FDA-approved (as Forzinity) for one rare disease; people use it off-label for energy and longevity.*
What it does	Mitochondria-targeted tetrapeptide - FDA accelerated approval as FORZINITY® for Barth syndrome (Sep 2025). Concentrates in the inner mitochondrial membrane via cardiolipin binding.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 0.25 mL	20 mg/mL	1 unit = 0.20 mg
10 mg	10mg + 0.50 mL	20 mg/mL	1 unit = 0.20 mg
50 mg	50mg + 2.5 mL	20 mg/mL	1 unit = 0.20 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-2	Daily AM	**2.5 mg**	12.5u	Tolerance ramp. Start low and assess gastrointestinal and injection-site response.
Weeks 3-12	Daily AM	**5 mg**	25u	Standard longevity-community working dose. FDA Forzinity is 40 mg/day SC for Barth syndrome only - that is the disease-modifying dose, not a longevity target.
Off	4-8 weeks	-	-	Consolidation. Sequences naturally into a MOTS-c cycle.
Cycle length	8-12 weeks on / 4 weeks off (longevity range). Max: ongoing under specialist supervision with mitochondrial monitoring (Forzinity is daily continuous for Barth).			
Timing	Daily AM or anytime — consistency more important than timing			

Handling & Stacking

Injection site	SubQ abdomen or outer thigh.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 4 weeks.
Side effects	Injection-site redness/swelling (most common). Headache, fatigue. Generally mild.
What to expect	Energy and exercise capacity improvement at 4-8 weeks. Injection-site redness expected. First step of the mitochondrial longevity sequence: SS-31 → MOTS-c →
Supply math	**5mg** → 76 vials | **10mg** → 38 vials | **50mg** → 8 vials *(12-wk @ 5 mg × 7/wk)*
Pairs well with	**MOTS-c** (mitochondrial stack - usually run as supportive precedent); **NAD+** (mitochondrial energy); Humanin (MDP pairing)
Avoid	Active mitochondrial disease without specialist supervision (FDA approved only for Barth syndrome under monitoring)
References	peptidedeck · realpeptides · medsbase · researchdosing · peptidedosages · DailyMed

Safety & Patient Education

Contraindications	• Pregnancy, lactation. • Active mitochondrial disease without specialist supervision.
Warnings	• FDA-approved (Forzinity) for Barth syndrome only; longevity use is off-label. • Cardiolipin binding saturates at ~1 mg/kg.
Labs	• Compound-specific: Fasting glucose (Week 0/4/8), CMP/BMP renal (baseline + mid-cycle) - renally cleared • Renal function (renally cleared)
Stop criteria	• Persistent injection-site reactions, headache not resolving
Patient education	• Aseptic technique: alcohol swab the vial septum and the injection site, use a single-use sterile needle for every injection, never share vials or needles between people • Sharps disposal in an FDA-cleared sharps container, never household trash • Cold chain: if a refrigerated peptide reaches room temperature for more than the labeled stability window, discard it • Mark the date of first puncture on every multi-dose vial • Inspect every vial before injection: clear solution (unless labeled otherwise), no particulates, no cloudiness, no color change • Rotate injection sites to avoid lipoatrophy, induration, and infection • Carry a record of what you're injecting if you need emergency care

TB-500 (THYMOSIN BETA-4)

TISSUE REPAIR AND RECOVERY

In plain English	*Whole-body repair peptide. Mobilizes stem cells, drops inflammation, and grows new blood vessels. Usually paired with BPC-157.*
What it does	Preclinical evidence (animal/cell-culture) only; no published human efficacy trials. Upregulates actin-binding via Tβ4-actin axis.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 1 mL	5 mg/mL	1 unit = 0.05 mg
10 mg	10mg + 2 mL	5 mg/mL	1 unit = 0.05 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-2	Daily	**250 mcg**	5u	Loading. Single injection near injury site; rotate.
Weeks 3-4	Daily	**375 mcg**	7.5u	Step-up. Half-unit increment optional - many users go directly to 10u/500 mcg.
Weeks 5-10	Daily	**500 mcg**	10u	Working dose. Matches Wolverine Blend TB-500 component.
Aggressive	1-2×/week	**4-6 mg**	80-120u	Community-standard loading bolus (researchdosing, peptidedosages). Used for acute soft-tissue injury, post-surgical recovery, or competitive-athlete recovery where rapid tissue saturation is the goal. 4-6 mg once or twice weekly for 4-6 weeks, then taper to the daily working dose or to maintenance 2 mg every 1-2 weeks.
Off	2-4 weeks	-	-	Standard washout. Re-cycle if continued tissue work needed.
Cycle length	8-10 weeks on / 2-4 weeks off (daily microdose, matches Wolverine). Aggressive bolus: 4-6 weeks loading / 4-8 weeks maintenance / 4 weeks off.			
Timing	Daily or M-F, anytime — Wolverine partner			

Handling & Stacking

Injection site	SubQ abdomen. Systemic action - injection site doesn't matter.
Storage	Lyophilised: -4°F (years stable). Reconstituted: 36-46°F, up to 8 weeks with continuous cold chain. Discard if any cloudiness or colour change.
Side effects	Head rush immediately post-injection (common, brief). Mild fatigue.
What to expect	Head rush immediately post-injection is normal. Systemic anti-inflammatory within days. Recovery improvements across all musculoskeletal tissues over 4-8 weeks.
Supply math	**5mg** → 6 vials | **10mg** → 3 vials *(10-wk @ 500 mcg × 7/wk)*
Pairs well with	**BPC-157** (**Wolverine stack** - joint, tendon, gut); **GHK-Cu** (skin healing + collagen); **Ipamorelin** / **CJC No DAC** (GH-mediated tissue repair); **Thymosin Alpha-1** (immune modulation)
Avoid	Active malignancy (promotes angiogenesis). Pre-existing tumour growth concern. **GLOW** / **KLOW Blend** (already contains **TB-500** - would double-dose)
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox · peptides.id

Safety & Patient Education

Contraindications	• Active malignancy (angiogenic), pregnancy, lactation
Warnings	• No published human efficacy trials — all use is off-label • Theoretical malignancy concern: tissue-repair angiogenesis can also support tumor vascularization
Drug interactions	• Anticoagulants / antiplatelets (warfarin, DOACs, aspirin, clopidogrel): caution due to vascular activity — no human data • Chemotherapy / anti-angiogenic drugs (bevacizumab, lenalidomide, sorafenib): mechanistically opposed — do NOT combine outside oncology supervision • VEGF-pathway agents: TB-500 activates angiogenesis — may oppose anti-VEGF therapy • BPC-157: complementary mechanism, commonly stacked — same active-malignancy contraindication
Labs	• Compound-specific: CBC + CMP baseline; no compound-specific markers routinely required
Stop criteria	• New unexplained mass, vascular changes (varicosities, persistent skin discoloration), persistent injection-site reactions
Patient education	• Research compound — no published human RCTs. All use is off-label; quality varies — request COA • Active cancer is an absolute contraindication — angiogenic mechanism can support tumor blood supply • Rotate injection sites; cycle 4-6 wks on / 2-4 wks off

TESAMORELIN

GROWTH HORMONE AXIS

In plain English	*Egrifta. FDA-approved peptide that shrinks belly (visceral) fat by triggering a GH pulse. Has to be done at night on an empty stomach.*
What it does	FDA-approved GHRH analog (Egrifta) - approved for HIV lipodystrophy at 2 mg/night. Full-length GHRH analog with trans-3-hexenoic acid modification. Binds GHRH receptors for strong nightly GH pulses.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 0.5 mL	10 mg/mL	1 unit = 0.10 mg
10 mg	10mg + 1 mL	10 mg/mL	1 unit = 0.10 mg
20 mg	20mg + 2 mL	10 mg/mL	1 unit = 0.10 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-16	Nightly	**2 mg**	20u	FDA Egrifta clinical dose (no titration required per label). Hold here for the full cycle.
Off	4-8 weeks	**-**	-	Pituitary and receptor recovery before re-cycling.
Cycle length	Ongoing per FDA Egrifta label (continuous nightly dosing for HIV-associated lipodystrophy). Community / off-label use sometimes cycles 12-16 weeks on / 4-8 weeks off with IGF-1 recheck - but the FDA-approved indication is continuous, not cycled.			
Timing	Pre-bed nightly, fasted. No food 90 min before; no food 60 min after — food within the window blunts GH response (per FDA Egrifta label)			

Handling & Stacking

Injection site	Abdomen only. Rotate quadrants.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, up to 28 days with continuous cold chain.
Side effects	Fluid retention, peripheral oedema, tingling. Glucose intolerance - monitor fasting glucose.
What to expect	Visceral fat reduction at 8-12 weeks. Strict night-only injection - food within 90 min blunts GH response significantly. Monitor fasting glucose.
Supply math	**5mg** → 44 vials | **10mg** → 22 vials | **20mg** → 11 vials *(16-wk @ 2 mg × 7/wk)*
Pairs well with	**CJC-1295 No DAC + Ipamorelin** (additive GH); **Tirzepatide** (visceral fat synergy)
Avoid	Active malignancy (FDA-labelled contraindication, USPI). Severe diabetic retinopathy. IGF-1 elevation should be monitored
References	accessdata.fda · rxlist · researchdosing · peptidedosages · DailyMed · PubMed

Safety & Patient Education

Contraindications	• Active malignancy of any kind (GH/IGF-1 can promote tumor growth) • Severe diabetic retinopathy • Acute critical illness (per somatropin label — applies to GH-axis stimulators) • Pregnancy and lactation
Warnings	• Insulin resistance and worsening glycemic control with sustained use • Fluid retention, peripheral edema • Carpal tunnel syndrome with extended use • Joint and muscle pain • IGF-1 elevation: theoretical malignancy promotion if dosed beyond physiological IGF-1 ranges • Combined GHRH+GHRP use requires off cycles to prevent receptor desensitization • Glucose intolerance and new-onset T2DM (boxed-equivalent in label) • Fluid retention with peripheral edema • Active malignancy contraindication is FDA-labeled (not just precautionary) • Severe diabetic retinopathy contraindication • Pituitary disease (hypopituitarism, post-surgical)
Drug interactions	• Glucocorticoids: blunt GH response — minimize or time away from injection • Insulin / antidiabetics: GH worsens insulin resistance — expect dose adjustments + tighter monitoring • Levothyroxine: GH lowers free T4 — recheck TSH/fT4 quarterly • Oral estrogens: blunt GH-stimulated IGF-1 — transdermal doesn't • CYP3A4 narrow-TI substrates (cyclosporine, tacrolimus, sirolimus): recheck troughs after starting
Labs	• Compound-specific: IGF-1 (baseline + Week 8 + end) - hold or reduce dose if IGF-1 >350-400 ng/mL, Fasting glucose (Week 0/4/8), Lipid panel (Week 0/12) • IGF-1 at baseline, mid-cycle, and end of cycle. Target ≤ULN for age • Fasting glucose and HbA1c at baseline and end of cycle • Mammogram and prostate screening per age-appropriate guidelines before starting • IGF-1 quarterly with target below 1× upper limit of normal for age
Stop criteria	• Fasting glucose rises >20 mg/dL above baseline • IGF-1 rises above the upper limit of normal for age • New persistent joint pain or carpal tunnel symptoms • New skin changes or moles enlarging • Vision changes • IGF-1 above 2x ULN, new-onset diabetes, persistent edema
Patient education	• GH-axis stimulation is not appropriate during active malignancy or for fertility • Most GH pulses occur during deep sleep — dose pre-bed unless instructed otherwise • Food within 90 min of injection blunts GH response (carbs/fats most) — fasted is standard • Report new lumps, persistent headaches, or vision changes immediately

THYMOSIN ALPHA-1

LONGEVITY AND BIOREGULATORS

In plain English	*An immune-system booster. Approved in 35+ countries (as Zadaxin) for chronic hepatitis. Not approved in the US.*
What it does	Master immune regulator - approved in 35+ countries for hepatitis, cancer, and immune deficiency. Promotes T-cell differentiation. Activates dendritic cells, NK cells, and cytotoxic T-lymphocytes. Modulates Th1/Th2 balance.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
5 mg	5mg + 1 mL	5 mg/mL	1 unit = 0.05 mg
10 mg	10mg + 2 mL	5 mg/mL	1 unit = 0.05 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Standard	2×/week	**1.6 mg**	32u	FDA Zadaxin label: 1.6 mg twice weekly, 3-4 days apart. Source previously read 3x/week.
Immune boost	Daily × 4 weeks	**1.6 mg**	32u	Intensive course.
Off	4 weeks	**-**	-	Essential break between cycles. Skipping leads to tolerance and diminishing returns.
Cycle length	6-12 weeks on, 4 weeks off.			
Timing	2×/week SubQ, 3-4 days apart (Zadaxin label)			

Handling & Stacking

Injection site	SubQ abdomen.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 4 weeks.
Side effects	Generally very well tolerated. Rare: injection-site reactions.
What to expect	Immune improvement over 4-12 weeks. CD4/CD8 ratio and NK counts are measurable endpoints. Very well tolerated - one of the safest immune peptides available.
Supply math	**5mg** → 4 vials | **10mg** → 2 vials *(6-wk @ 1.6 mg × 2/wk)*
Pairs well with	**CJC-1295 No DAC + Ipamorelin** (recovery + immune stack); **BPC-157** / **TB-500** (tissue repair); **Epithalon** (longevity); **NAD+** (mitochondrial immune support); **Glutathione** (antioxidant); **GHK-Cu** (skin/immune repair)
Avoid	Active autoimmune flare (immune amplification may worsen). Severe immunosuppression therapy (interaction)
References	researchdosing · peptidedosages · PubMed · thepeptidereport · peptidefox · peptides.id

Safety & Patient Education

Contraindications	• Active autoimmune flare, severe immunosuppression therapy
Warnings	• FDA placed thymosin alpha-1 on the 2023 503A "may not be compounded" list (verify current status). • Approved as Zadaxin in 35+ countries but not in the US.
Drug interactions	• Immunosuppressants (cyclosporine, tacrolimus, sirolimus, mycophenolate, methotrexate): pharmacologically opposed — Tα1 augments T-cell function while these suppress it. Combining is generally inappropriate outside a specialist-supervised protocol • Corticosteroids (chronic systemic): blunt the Tα1 immune response; coordinate dosing or pause steroids during a Tα1 course where clinically possible • Live vaccines: theoretical augmented response — coordinate timing with the prescriber • TNF-alpha inhibitors and other biologics (adalimumab, etanercept, infliximab, rituximab): mechanistically opposed for autoimmune indications; not advisable to combine without specialist oversight • Antiviral therapy for hepatitis B/C: Tα1 is studied as adjunct (Zadaxin label uses it that way) — synergistic, not antagonistic
Labs	• Compound-specific: T-cell panel CD3/CD4/CD8/NK (baseline + post-course) • T-cell panel (CD3/CD4/CD8/NK)
Stop criteria	• Autoimmune flare, new immune-mediated symptoms
Patient education	• Used clinically as an adjunct to antiviral therapy for chronic hepatitis B and C in countries where Zadaxin is approved. • Cyclic dosing (typically 1.6 mg SubQ 2× weekly for 4-12 weeks) is the canonical pattern; continuous dosing isn't supported by the literature. • Quality and authenticity vary widely in the US compounding market — verify supplier and request COA. • Stop and contact a clinician for any new joint pain, rash, fevers, or symptoms suggesting an autoimmune flare.

TIRZEPATIDE

FAT LOSS AND METABOLISM

In plain English	*Mounjaro and Zepbound. Weekly shot that hits two appetite hormones at once. Got people up to ~22% weight loss at the high dose.*
What it does	FDA-approved dual GIP/GLP-1 agonist - gold standard for weight loss. Co-activates GIPR (appetite + insulin + fat cell effects) and GLP-1R (satiety + gastric slowing + glucagon suppression).

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 0.50 mL	20 mg/mL	1 unit = 0.20 mg
15 mg	15mg + 0.75 mL	20 mg/mL	1 unit = 0.20 mg
20 mg	20mg + 1 mL	20 mg/mL	1 unit = 0.20 mg
30 mg	30mg + 1.5 mL	20 mg/mL	1 unit = 0.20 mg
40 mg	40mg + 2 mL	20 mg/mL	1 unit = 0.20 mg
50 mg	50mg + 2.5 mL	20 mg/mL	1 unit = 0.20 mg
60 mg	60mg + 3 mL	20 mg/mL	1 unit = 0.20 mg
80 mg	80mg + 4 mL	20 mg/mL	1 unit = 0.20 mg
100 mg	100mg + 5 mL	20 mg/mL	1 unit = 0.20 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-4	1× weekly	**2.5 mg/week**	12.5u	FDA standard starting dose. Hold here for 4 weeks before stepping up; most GI side effects occur early.
Weeks 5-8	1× weekly	**5 mg/week**	25u	Step up only if Weeks 1-4 tolerated. Many users find sufficient appetite control here.
Weeks 9-12	1× weekly	**7.5 mg/week**	37.5u	Common maintenance dose. Many users (including strong responders) stop here long-term.
Weeks 13-16	1× weekly	**10 mg/week**	50u	Step up only if 7.5 mg appetite control is fading. If 7.5 mg is still working, no reason to climb.
Weeks 17-20	1× weekly	**12.5 mg/week**	62.5u	Reserved for users who need more. Side effects are dose-dependent - higher dose is not better if a lower one is working.
Weeks 21+	1× weekly	**15 mg/week**	75u	FDA Mounjaro/Zepbound maximum. Most users never need this. Stop at any earlier rung that controls appetite without intolerable side effects.
Cycle length	Ongoing per FDA Mounjaro / Zepbound labels. No scheduled off-cycle. Discontinue at least 1 month before planned pregnancy. Compounding note: after March 19, 2025, FDA bars compounded tirzepatide without documented clinical difference - sourcing now mostly via grey-market research suppliers.			
Timing	Weekly (any day) — same day each week; time of day flexible			

Handling & Stacking

Injection site	Abdomen, outer thigh, or upper arm.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, use within 28 days.
Side effects	Nausea, vomiting, diarrhea, constipation. Generally better tolerated than semaglutide.
What to expect	Appetite reduction from week 1. Generally better GI tolerance than semaglutide. ~20% average weight loss at 72 weeks. Plateau-breaking at higher doses common.
Supply math	**50mg** → 6 vials | **60mg** → 6 vials | **80mg** → 4 vials | **100mg** → 4 vials *(21-wk @ 15 mg × 1/wk)*
Pairs well with	**Tesamorelin** (visceral fat focus, GH support); **5-Amino-1MQ** (NNMT inhibition, complementary metabolic pathway)
Avoid	**Semaglutide** or **Retatrutide** simultaneously, or any other GLP-1 receptor agonist (additive nausea, GLP-1 desensitisation). Insulin without close glucose monitoring (hypoglycaemia risk)
References	glp3planner · seekpeptides · rxlist · researchdosing · peptidedosages · DailyMed

TIRZEPATIDE

FAT LOSS AND METABOLISM — SAFETY & PATIENT EDUCATION

Contraindications	• Personal or family history of medullary thyroid carcinoma (MTC) or MEN 2 • Personal or family history of any thyroid cancer • Personal history of pancreatitis (any cause) • Pre-existing gastroparesis or significant gastric motility disorder • Severe GERD, scleroderma, or other conditions with delayed gastric emptying • Prior serious hypersensitivity to the compound or excipients • Pregnancy and lactation; discontinue ≥2 months (sema) or ≥1 month (tirz) before planned conception • History of cholelithiasis or gallbladder disease • Active or pre-existing diabetic retinopathy • Eating disorder history (rapid weight loss can trigger relapse) • Chronic kidney disease stage 4-5 • History of any cancer (slowed gastric motility complicates oral chemotherapy absorption)
Warnings	• Acute kidney injury from GI-related volume depletion — risk highest during dose escalation • Acute pancreatitis: discontinue if suspected (severe abdominal pain radiating to back) • Acute gallbladder disease (cholelithiasis, cholecystitis) • Diabetic retinopathy progression in patients with pre-existing retinopathy • Hypoglycemia with insulin or sulfonylureas — reduce those at GLP-1 initiation • Pulmonary aspiration during anesthesia. Hold ≥1 wk (weekly) or day-of (daily) per ASA • Severe GI: ileus, obstruction, severe constipation, fecal impaction • Anaphylaxis and angioedema reported postmarketing • Tachycardia: small mean HR increases (1-4 bpm) reported
Drug interactions	• Delayed gastric emptying affects oral absorption (levothyroxine, oral contraceptives, warfarin) — separate timing • Insulin / secretagogues: dose reduction at GLP-1 initiation • Other GLP-1 agonists: do not combine
Pregnancy / Lactation	• Semaglutide: stop ≥2 mo before planned pregnancy (long half-life). Tirzepatide: stop ≥1 mo per Mounjaro label.
Labs	• Compound-specific: BMP/CMP (creatinine, eGFR - baseline + dose escalation), Fasting glucose (Week 0/4/8), HbA1c (baseline + end), Lipid panel (Week 0/12), Lipase (if abdominal pain) • Baseline: BMP/CMP, CBC, HbA1c, lipid panel, TSH, pregnancy test (childbearing potential) • During therapy: BMP at every dose escalation, and any time GI side effects cause >24h reduced food/fluid intake • HbA1c at 3 + 6 mo; lipid panel at 6 mo; repeat eGFR if signs of dehydration • Drop hsCRP and routine calcitonin (not standard of care)
Stop criteria	• Creatinine rise >0.3 mg/dL or eGFR drop >20%: hold, rehydrate, reassess • Severe abdominal pain: hold and evaluate for pancreatitis • Vision changes: ophthalmology evaluation • Persistent vomiting >48h: hold dose • New neck mass, hoarseness, dysphagia, dyspnea: thyroid pathology workup
Patient education	• AKI on GLP-1s is from dehydration, not the drug. Target 2.5-3 L water/day; add electrolytes if low PO. Hold next dose for any day of vomiting/diarrhea/reduced food; resume same level after 24h recovery. Same-day contact for: decreased urination, dark urine, flank pain, ankle swelling, confusion. • Required during any GLP-1 cycle: protein ≥1.2 g/kg IBW/day (1.6 if training), resistance training 2-3×/wk, creatine 5 g/day. Without this, 25-40% of weight lost is lean mass and regain is rapid. • Eat slowly, smaller portions, stop at first fullness • Same-day medical contact: severe abdominal pain (especially radiating to back), persistent vomiting, decreased urination, dark urine, vision changes, severe constipation, neck mass, hoarseness • Inform any anesthesiologist before procedures (aspiration risk) • Use contraception during therapy and washout • Pen sharing prohibited even with new needle • Discontinue at least 1 month before planned pregnancy (per Mounjaro label, shorter washout than semaglutide because shorter half-life). • Pen sharing prohibited.

VITAMIN B12

SUPPLEMENTS AND COFACTORS

In plain English	*The vitamin you can't get from plants. Injected for energy, mood, and methylation — especially useful for vegans, people on metformin, and anyone over 60.*
What it does	Essential B-vitamin for nerve function, red blood cell formation, and energy metabolism. Cofactor for methionine synthase (methylation) and methylmalonyl-CoA mutase (energy metabolism).

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 2.5 mL	4 mg/mL	1 unit = 0.04 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Deficiency loading	Daily ×7 then weekly	**1,000 mcg**	25u	Target serum B12 400-900 pg/mL.
Maintenance	Monthly	**1,000 mcg**	25u	Once repleted.
MIC protocol	M/W/F	**1 mg/mL in blend**	25u	As component of lipotropic blend.
Cycle length	monthly maintenance ongoing. Loading: 1000 mcg daily x 7 days then weekly x 4 weeks for deficiency. No cycling required for maintenance.			
Timing	Weekly IM (maintenance); daily for repletion phase			

Handling & Stacking

Injection site	SubQ or IM (IM preferred for deficiency repletion).
Storage	Solution: room temp or refrigerated. Use within 30 days. Protect from light.
Side effects	Rare. Mild injection-site irritation.
What to expect	Energy improvement within days of loading in deficient individuals. Neurological symptoms may improve over 4-8 weeks. Target serum B12: 400-900 pg/mL.
Supply math	**10mg** → 1 vial *(4-wk @ 1 mg × 3/wk)*
Pairs well with	**Lipo-C** (lipotropic stack); **NAD+** (cellular energy); GLP-1 agonists (counters **B12** depletion)
Avoid	Polycythaemia vera (**B12** amplifies erythrocyte production). Cobalt allergy
References	researchdosing · peptidedosages · DailyMed · PubMed · ClinicalTrials · Drugs.com

Safety & Patient Education

Contraindications	• Polycythemia vera, cobalt allergy, Leber's optic neuropathy (cyanocobalamin form specifically)
Warnings	• Methylcobalamin preferred over cyanocobalamin for most uses
Drug interactions	• Proton pump inhibitors (omeprazole, pantoprazole, esomeprazole) and H2 blockers: long-term use reduces oral B12 absorption; injectable B12 bypasses this and is the preferred route in PPI users • Metformin: chronic use depletes B12; check B12 + MMA annually in metformin users • Neomycin and other oral aminoglycosides: reduce intestinal B12 absorption; injectable route bypasses • Colchicine, chloramphenicol: interfere with B12 absorption or response to B12 therapy • Nitrous oxide: inactivates B12; avoid recreational or repeat anesthetic exposure during deficiency treatment • Folate supplementation: high-dose folate can mask B12 deficiency on CBC while neurological deterioration continues — always check both before high-dose folate
Labs	• Compound-specific: CBC (baseline - confirm macrocytic anemia resolution), Serum B12 (baseline + Week 8; target 400-900 pg/mL); MMA + homocysteine if neurological symptoms • B12, MMA, homocysteine baseline
Patient education	• Cyanocobalamin contraindicated in Leber's; use methylcobalamin or hydroxocobalamin

WOLVERINE BLEND

BLENDS AND MIXED PROTOCOLS

In plain English	*The Wolverine Blend. Two of the most popular healing peptides pre-mixed in one vial.*
What it does	Wolverine Blend (BPC-157 + TB-500) - pre-mixed healing blend: BPC-157 (local repair + angiogenesis) + TB-500 (systemic actin modul BPC-157 activates VEGF/NO/GHR pathways for local tissue repair.

Reconstitution

Vial size	BAC water / Solvent	Concentration	Note
10 mg	10mg + 1 mL	10 mg/mL	1 unit = 0.10 mg
20 mg	20mg + 2 mL	10 mg/mL	1 unit = 0.10 mg
30 mg	30mg + 3 mL	10 mg/mL	1 unit = 0.10 mg

Protocol

Weeks	Frequency	Dose	Units	Notes
Weeks 1-2	Daily	**500 mcg**	5u	Loading: assess tolerance. Single injection sufficient at this dose.
Weeks 3-4	Daily	**750 mcg**	7.5u	Step up only if Weeks 1-2 well tolerated. Half-unit step is optional - many users go directly to 1 mg.
Weeks 5-10	Daily AM	**1 mg**	10u	Working dose. Once daily preferred; optional AM/PM split (5u + 5u) if you find injection volume uncomfortable.
Aggressive	Twice daily (AM + PM split)	**1 mg × 2**	10u	Community-aggressive for acute soft-tissue injury, post-surgical recovery, or stalled response after 6+ weeks at working dose. 1 mg blend BID = 1 mg of each component daily (matches BPC-157 standalone Aggressive). 4-6 weeks then taper to the daily working dose.
Off	2-4 weeks	-	-	Standard washout. Re-cycle if continued tissue work needed.
Cycle length	8-10 weeks on / 2-4 weeks off (daily microdose). Aggressive (BID acute injury): 4-6 weeks split-dose / 4 weeks off, then back to daily working dose.			
Timing	Daily, anytime — Wolverine Blend; food doesn't affect			

Handling & Stacking

Injection site	SubQ near injury site. Rotate.
Storage	Lyophilised: -4°F. Reconstituted: 36-46°F, up to 8 weeks with continuous refrigeration.
Side effects	Usually minimal. Occasional lightheadedness.
What to expect	Faster recovery than either alone. Pain reduction within days, structural repair over 4-12 weeks. Systemic anti-inflammatory effect complements localised healing.
Supply math	**10mg** → 6 vials | **20mg** → 3 vials | **30mg** → 2 vials *(10-wk @ 1 mg × 7/wk)*
Pairs well with	**GHK-Cu** (skin + collagen synergy); **Ipamorelin** / **CJC No DAC** (GH-mediated repair); **Tesamorelin** (GHRH adjunct); **KPV** (gut anti-inflammatory)
Avoid	NSAIDs (antagonise prostaglandin pathway). Active malignancy. High-dose vitamin C same session. **GLOW** / **KLOW Blend** (already contains **BPC** + TB - would double-dose)
References	peptidefox · researchdosing · peptidedosages · PubMed · thepeptidereport · peptides.id

Safety & Patient Education

Contraindications	• Active malignancy of any kind (TB-500 angiogenic; BPC-157 growth-promoting) • Active bleeding or recent hemorrhagic stroke (TB-500 vascular) • Pregnancy, lactation
Warnings	• No published human RCTs for the combination • BPC-157: FDA Category 2 (2023) — compounding from bulk substances restricted • TB-500: theoretical malignancy concern via angiogenesis; no human efficacy trials • Vial quality varies — request COA for each lot • Tissue-repair pathways overlap with cancer — any unexplained mass requires evaluation
Drug interactions	• NSAIDs: rodent data suggests blunted BPC-157 effect — clinical relevance uncertain in humans • Anticoagulants / antiplatelets (warfarin, DOACs, aspirin, clopidogrel): caution with TB-500 vascular effects — monitor INR/anti-Xa if combining • Chemotherapy: do not combine — both promote angiogenesis/tissue growth • Other peptides at same site: rotate sites to reduce local irritation
Labs	• Compound-specific: hsCRP (baseline + end)
Stop criteria	• New mass or unexpected tissue growth at any injection site • Persistent injection-site reactions >7 days (redness, induration, abscess) • New unexplained GI symptoms or bleeding • Vascular changes — new varicosities, persistent skin discoloration
Patient education	• Two-peptide stack — any single-component contraindication applies. Active cancer is an absolute stop • Both are research peptides without published human RCTs. All use is off-label; quality varies • Local injection near target tissue for orthopedic use; SubQ near abdomen for systemic. Rotate sites • Cycle: 4-6 wks on / 2-4 wks off. Continuous dosing isn't supported and may blunt response • Stop and contact clinician for new lump, persistent injection-site reaction >7 days, or GI bleeding

Sources & References

Sites consulted while researching, cross-auditing, and verifying the protocols in this guide.

Tier 1 — Community protocol databases (heavy use)

researchdosing.com — per-vial dosing pages; primary protocol source. Referrer answer: Certified Peptides. Used for HCG, GHK-Cu, Tesamorelin, CJC+Ipa, 5-Amino-1MQ, SS-31, MT-2, MOTS-c, Semax, GLOW, KLOW, Retatrutide, Wolverine.
peptidedosages.com — single-peptide and blend protocol pages with supply math. Covers NAD+ (500mg + 1000mg), GHK-Cu (50mg + 100mg), Glutathione, BPC+TB Wolverine, CJC+Ipa, Tesamorelin, 5-Amino-1MQ, Epithalon, Pinealon.
thepeptidereport.com — per-peptide protocol pages. Used for Tesamorelin, GHK-Cu, MOTS-c, Glutathione, NAD+, Semax.
peptidedosingprotocols.com — aggregator with FDA cross-references. Used for GHK-Cu, Tesamorelin, CJC+Ipa stack, Wolverine stack, MOTS-c, NAD-Plus, SS-31, Semax.
peptidefox.com — blend protocols, injection technique, peptide calculator. Wolverine stacking guide, GLOW/KLOW protocol, BPC-157, Retatrutide dual-axis stack.
peptides.id — research-purpose dosing schedules and mechanism summaries (non-FDA compounds). Note: page structure not crawlable for automated retrieval — manually referenced.

Tier 2 — Pharmacy / clinic protocols

empowerpharmacy.com — NAD+ injection, Bremelanotide compounded protocols
olympiapharmacy.com — NAD dosage charts (per-day, per-week)
agelessrx.com — NAD+ injection clinical overview
fagronacademy.us — NAD+ pharmacy review article
defymedical.com — PT-141, HCG TRT-adjunct protocols
walnutcreekaesthetics.com — NAD+ injection SOP
jinfiniti.com — NAD+ dosage chart
tailormadecompounding.com — research-peptide compounding pharmacy. Note: public catalog gated behind clinician login — not retrievable, only catalog summaries used.

Tier 3 — FDA / clinical primary sources

dailymed.nlm.nih.gov — FDA prescribing info: Pregnyl, Novarel, Ovidrel, Egrifta SV/WR, Vyleesi, Forzinity
accessdata.fda.gov — FDA drug-approval database (USPI labels — Vyleesi NDA 210557 and others)
pubmed.ncbi.nlm.nih.gov — BPC-157 IV safety pilot, KPV PepT1 uptake, Selank, Semax, Elamipretide TAZPOWER, MOTS-c mechanism
pmc.ncbi.nlm.nih.gov — Retatrutide TRIUMPH Phase 3 (PMC12673447), MOTS-c review, Elamipretide failing heart, Semax brain ischemia
clinicaltrials.gov — Retatrutide NCT05929066 (TRIUMPH-1) and related trial registry entries
ema.europa.eu — European Medicines Agency drug-approval database. Searched for EU labels on Tesa/Egrifta and GLP-1s. Note: no EU label retrieved — Tesamorelin withdrawn from EU 2012; GLP-1s use EU-specific brand names.

Tier 4 — Reference / encyclopedic

en.wikipedia.org — HCG, MOTS-c, Semax, Elamipretide, Tesamorelin, BPC-157, Bremelanotide, GHK-Cu, CJC-1295, Ipamorelin, Melanotan II
drugs.com — Tesamorelin/Egrifta, Vyleesi, Forzinity, HCG, Retatrutide monographs
rxlist.com — Choriogonadotropin alfa, Ovidrel, Pregnyl, Novarel, Egrifta, Elamipretide, Semaglutide

Tier 6 — Other protocol-adjacent

peptidedeck.com — GHK-Cu dosage guide, Elamipretide review, side-effect data
seekpeptides.com — Tirzepatide reconstitution chart, GHK-Cu dosing
perfectb.com — Epithalon protocol, GHK-Cu 30-day cycle
glp3planner.com — Tirzepatide reconstitution guide

Government / educational technique sources

cdc.gov — Subcutaneous injection technique (Pink Book Chapter 6)
ncbi.nlm.nih.gov/books — Injection best practices (NBK138495)

Aesthetic / clinic / consumer (one-off references)

my.clevelandclinic.org — HCG
emedicine.medscape.com — HCG reference range
go.drugbank.com — HCG DrugBank entry
aboutnad.com / peachiv.com / thinworks.com / hubmeded.com / atlasmenshealth.com — NAD+ consumer pages
innerbody.com / klearmindclinics.com / prospecbio.com / exploring-peptides.com / rwacenter.com — Semax pages
peptides.org / peptidescalculator.com / peptide-db.com / dosagepeptide.com — Calculators, dose info
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medsbase.com — SS-31 guide
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curefa.org — Forzinity drug development
peptideinitiative.com — Pinealon, GHK-Cu
elementsarms.com / trimrx.com / pinnaclepeptides.com / tirzepatidepeptide30mg.com / tirzepatidedosagecalculator.com / calorize.com.ua — Tirzepatide pages
mypeptidematch.com — Glutathione
cellgenic.com — Peptide calculator

Community discussion (not used for protocol verification)

reddit.com/r/Peptides — Subreddit search for dosing reports and sourcing experience. Intentionally not retrieved for primary protocol verification — anecdotal, not credibility-weighted.